

# Laboratory Manual

## **Introduction to Communication Engineering Laboratory**

**EEE/ECE 342 (V1, V2)**

**EEE/ECE 341L (V3)**

**Department of Electrical & Electronic Engineering (EEE)  
School of Engineering (SoE)  
BRAC University**



**Revision: June 2026**

### A. Course Objectives:

The objectives of this course are to:

- Implement the core concepts and fundamental elements of a communication system by means of hardware setup and software.
- Provide students with sound understanding and knowledge of different modes of modulation schemes used in modern communication systems and basic multiplexing techniques through practical demonstration.

### B. Course Outcomes, CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

| Sl.                                                                      | CO Description                                                                               | POs | Bloom's taxonomy domain/level            | Delivery methods and activities | Assessment tools                |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----|------------------------------------------|---------------------------------|---------------------------------|
| EEE_ECE 341 Introduction to Communication Engineering                    |                                                                                              |     |                                          |                                 |                                 |
| CO1                                                                      | <b>Explain</b> different types of modulation and multiplexing techniques.                    | a   | Cognitive/ Understand                    | Lectures, Notes                 | Assignment, Quiz, Exam          |
| CO2                                                                      | <b>Analyze</b> modulated and demodulated signals in time domain and frequency domain.        | b   | Cognitive/ Analyze                       | Lectures, Notes                 | Assignment, Exam                |
| CO3                                                                      | <b>Apply</b> the knowledge to solve problems related to communication engineering.           | a   | Cognitive/ Apply                         | Lectures, Notes                 | Assignment, Quiz, Exam, Project |
| EEE_ECE 341L / EEE_ECE 342 Introduction to Communication Engineering Lab |                                                                                              |     |                                          |                                 |                                 |
| CO4                                                                      | <b>Use</b> hardware and software tools to perform experiments on various modulation schemes. | e   | Cognitive/ Apply/ Psychomotor/ Precision | Lab Class                       | Lab Work, Lab Exam              |

### C. Mark Distribution

| Assessment Tools  | Weightage (%) |
|-------------------|---------------|
| Attendance        | 10            |
| Class Performance | 10            |
| Lab Report        | 20            |
| Hardware Exam     | 20            |
| Project           | 25            |
| Lab Final (Quiz)  | 15            |

### D. References

| Sl. | Title                                           | Author(s) | Publication | Edition                 | Publisher | ISBN              |
|-----|-------------------------------------------------|-----------|-------------|-------------------------|-----------|-------------------|
| 1   | Modern Digital and Analog Communication Systems | B. Lathi  | 2006        | 3 <sup>rd</sup> edition | Oxford    | 978-0-19-511009-8 |
| 2   | Communication Systems                           | S. Haykin | 2005        | 4 <sup>th</sup> edition | Wiley     | 978-0-47-117869-9 |

## **Lab Safety and Security Issues**

**(Please modify this part as you feel appropriate for your lab)**

### **1. Laboratory Safety Rules (General Guidelines):**

The Department of EEE maintains general safety rules for laboratories. The guideline is attached in front of the door in each of the laboratories. The written rules are as follows.

- A. Closed shoes must be worn that will provide full coverage of the feet and appropriate personnel clothing must be worn.
- B. Always check if the power switch is off before plugging in to the outlet. Also, turn the instrument or equipment OFF before unplugging from the outlet.
- C. Before supplying power to the circuit, the connections and layouts must be checked by the teacher.
- D. Voltages equal or above 50V are always dangerous. Therefore, extra precautions must be taken as voltage level is increased.
- E. Extension cords should be used only when necessary and only on a temporary basis.
- F. Once the lab exercise is done, all equipment must be powered down and all probes, cords and other instruments must be returned to their proper position.
- G. In case of fire, disconnect the electrical mains power source if possible.
- H. Students must be familiar with the locations and operations of safety and emergency equipment like Emergency power off, Fire alarm switch and so on.
- I. Eating, drinking, chewing gum inside electrical laboratories are strictly prohibited.
- J. Do not use damaged cords or cords that become too hot or cords with exposed wiring and if something like that is found, inform the teacher/LTO right away.
- K. No laboratory equipment can be removed from their fixed places without the teacher/LTO's authorization.
- L. No lab work must be performed without the laboratory teacher/lab technical officer being present.

### **2. Electrical Safety**

To prevent electrical hazards, there are symbols in front of the Electrical Distribution Board, High voltage three phase lines in the lab, Backup generator and substation. Symbols related to Arc Flash and Shock Hazard, Danger: High Voltage, Authorized personnel Only, no smoking etc. are posted in required places. Only authorized personnel are allowed to open the distribution boxes.

### **3. Electrical Fire:**

If an electrical fire occurs, try to disconnect the electrical power source, if possible. If the fire is small, you are not in immediate danger, use any type of fire extinguisher except water to extinguish the fire. When in doubt, push in the Emergency Power Off button.

### **4. IMPORTANT:**

Do not use water on an electrical fire.

### Lab Plan (Week-based)

| Serial | Experiment Name                                                                                                  | Page  | Tentative schedule | Is this experiment used for any CO assessment? |    |
|--------|------------------------------------------------------------------------------------------------------------------|-------|--------------------|------------------------------------------------|----|
|        |                                                                                                                  |       |                    | Yes                                            | No |
| 1      | Introductory Session: Discussion regarding Course material, Course Outcomes, Marks Distribution, Assessment Plan | 5     | Week 1             |                                                | No |
| 2      | Experiment 1: Introduction to MATLAB Simulink for Communication Systems                                          | 6-11  | Week 2             |                                                | No |
| 3      | Experiment 2: Double Sideband (DSB) Amplitude Modulation (AM) Generation                                         | 12-15 | Week 3             | Yes                                            |    |
| 4      | Experiment 3: Single Sideband (SSB) AM Generation & Detection                                                    | 16-21 | Week 4             | Yes                                            |    |
| 5      | Experiment 4: Frequency modulation and demodulation (Hardware)                                                   | 32-46 | Week 5             | Yes                                            |    |
| 6      | Experiment 5: Sampling and Signal Reconstruction (Hardware)                                                      | 47-54 | Week 6             | Yes                                            |    |
| 7      | Theory Mid Exam break                                                                                            |       | Week 7             |                                                |    |
| 8      | Experiment 6: Amplitude Shift Keying (ASK)                                                                       | 61-64 | Week 8             | Yes                                            |    |
| 9      | Experiment 7: Frequency Shift Keying (FSK)                                                                       | 65-70 | Week 9             | Yes                                            |    |
| 10     | Experiment 8: Phase Shift Keying (PSK)                                                                           | 71-75 | Week 10            | Yes                                            |    |
| 11     | Lab Quiz                                                                                                         |       | Week 11            | Yes                                            |    |
| 12     | Hardware Exam                                                                                                    |       | Week 12            | Yes                                            |    |
| 13     | Project Presentation and Report Submission                                                                       | 73-75 | Week 13            | Yes                                            |    |

**Updated by:**

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2. Saif Kabir

## ***Introductory Session: Discussion regarding Course material, Course Outcomes, Marks Distribution, Assessment Plan***

During the lab session of the first week the following topics should be discussed:

- Introducing the course materials, course outcomes, marks distribution, and assessment plan
- Instructions regarding group formation
- Lab report submission guidelines
- Software Installation procedures

### **1. Introducing the course materials, course outcomes, marks distribution, and assessment plan:**

The list of experiments should be provided to the students, and also an overview of the topics can be given. The course outcomes should be explained so that the students can have a clear understanding of what they are expected to learn from this course. The complete marks distribution and the assessment plan should be discussed as well.

### **2. Instructions regarding group formation:**

The following instructions could be provided to the students for the group formation purpose:

- a. Each group should have at most four members (exceptions can be made depending upon the available hardware facility).
- b. A group should be formed with the students having the same course code.
- c. All group members of a group should belong to the same section.

### **3. Lab report submission guideline:**

General guidelines for submitting the report could be as follows:

- a. Each group will submit one report per experiment
- b. The hard copy of the report should be submitted before the next lab session (a soft copy should also be submitted on the online platform, if necessary).
- c. All group members should contribute equally. If any member is not cooperating, you may remove his/her name from the lab report and send an email to the instructor informing the issue.
- d. Late submission will not be accepted.
- e. Plagiarism will be treated harshly.
- f. A cover page should be added with the report including the course code, course name, experiment no., experiment name, name, and ID of each of the group members.
- g. The report would be assessed in terms of comprehensiveness, cleanliness, and overall presentation.



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**Department of Electrical and Electronic Engineering**  
**EEE/ECE341L/342: Introduction to Communication Engineering**  
**Laboratory**

Experiment No: 1 (Software)

Experiment Name: - Introduction to MATLAB Simulink for Communication Systems

**Objectives:**

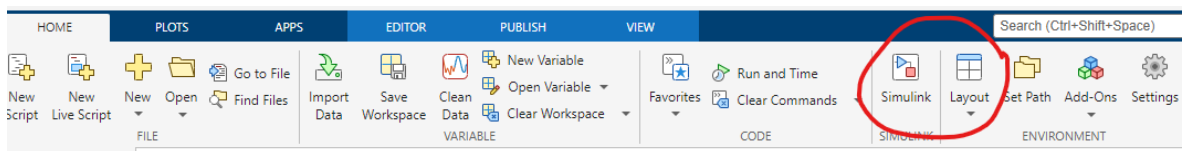
1. To familiarize with MATLAB Simulink tools for communication systems
2. To familiarize with different toolboxes for communication systems

**Equipment:**

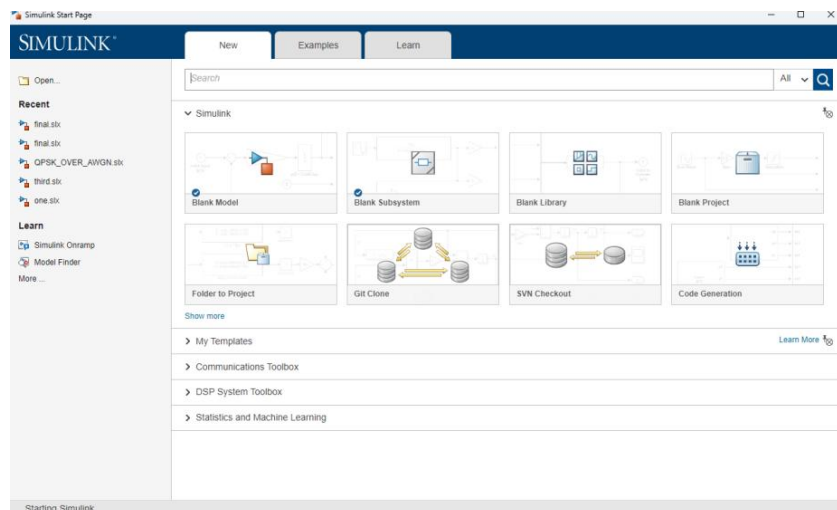
- 1) MATLAB/Simulink

**Step by Step Guidelines**

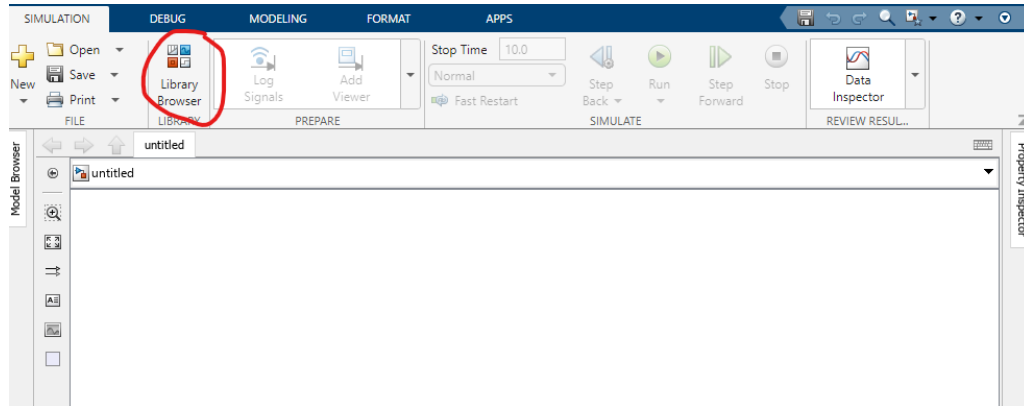
1. Start MATLAB and open Simulink by typing 'Simulink' in the command window or clicking the Simulink icon in the top menu.



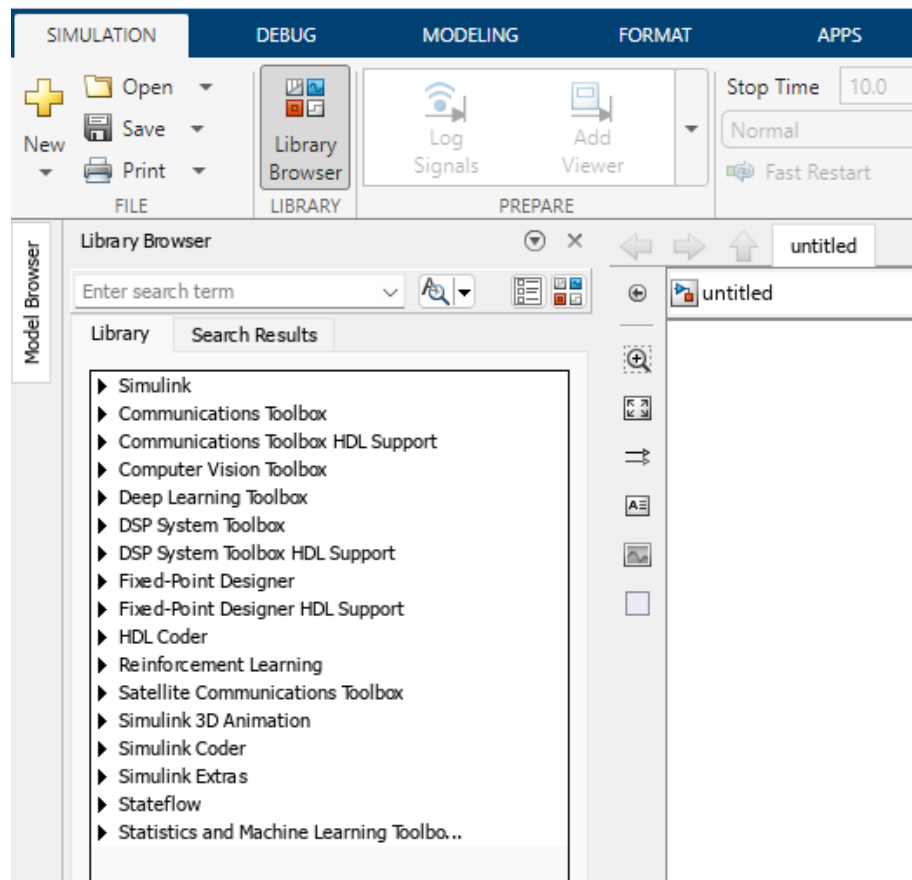
2. In Simulink, select 'Blank Model' to start a new simulation.



3. Click on the 'Library Browser' icon (four colored squares) at the top of the working area.



4. Browse through the library to explore available blocks. Focus on blocks related to signal processing and communication systems.



5. Identify and explore the following block categories:
  - a. Simulink/Commonly Used Blocks
  - b. Simulink/Math Operations
  - c. Simulink/Sinks
  - d. Simulink/Sources

- e. DSP System Toolbox/Math Operations
  - f. DSP System Toolbox/Signal Management
  - g. DSP System Toolbox/Sinks
  - h. DSP System Toolbox/Sources
  - i. DSP System Toolbox/Transforms
- 

### Example 1: Creating a Sine Wave Simulation

1. Open Simulink and create a Blank Model.
2. From the Library Browser, drag a 'Sine Wave' block from Simulink/Sources into the model.
3. Drag a 'Scope' block from Simulink/Sinks into the model.
4. Connect the output of the Sine Wave block to the input of the Scope block.
5. Run the simulation to visualize the sine wave in the time domain.

#### Building a System

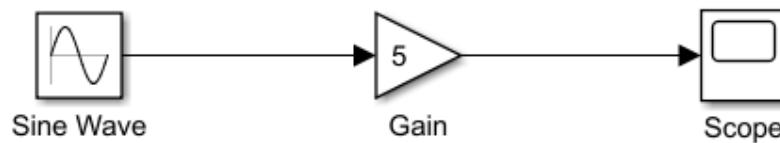


Figure 1: Simulation of a Sine wave

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### Example 2: Simulating a Spectrum Analyzer

**Goal:** Visualize signals in both time and frequency domains using Simulink blocks.

#### Steps:

1. Time Domain: Use the built-in Scope block.
2. Frequency Domain: Build a custom spectrum analyzer using:
  - Buffer (Size: 1024, No overlap)
  - Magnitude FFT (Length: 1024)
  - Array Plot
3. Signal Source: Use DSP Sine Wave (e.g., 10.5 kHz, 1 Vp).
4. Connect the blocks as shown in Figure 2(a) (a path with Buffer → FFT → Array Plot).
5. Use another branch to connect the same sine source directly to the built-in Spectrum Analyzer for comparison.
6. Visualization Tips:
  - Use "View → Style → Marker" to show data points.
  - Experiment with buffer/FFT lengths (128–4096, powers of 2).

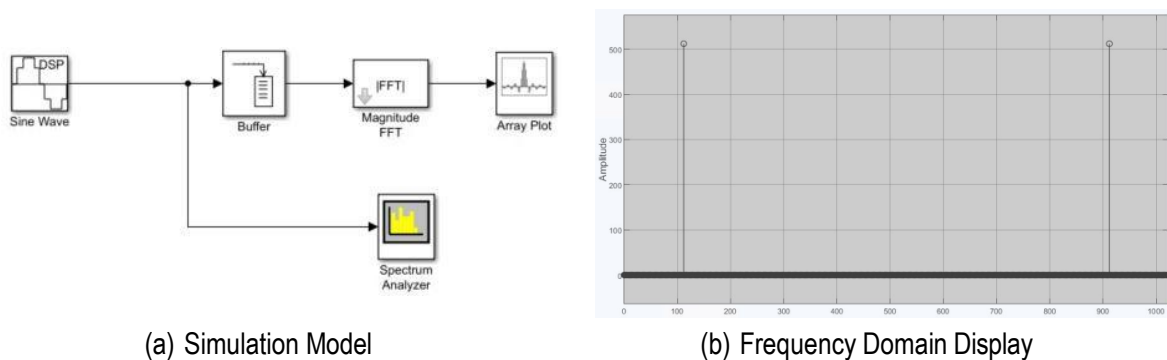


Figure 2. A Model for Visualizing Time and Frequency Domain

Figure 3 below shows the display of the Spectrum Analyzer. A few details should become apparent: the y-axis in the plot is displayed in dBm, the x-axis displays Hertz (positive AND negative), and the plot presents noise. Take your time now to search *dBm*, *negative frequency* and *noise floor*.

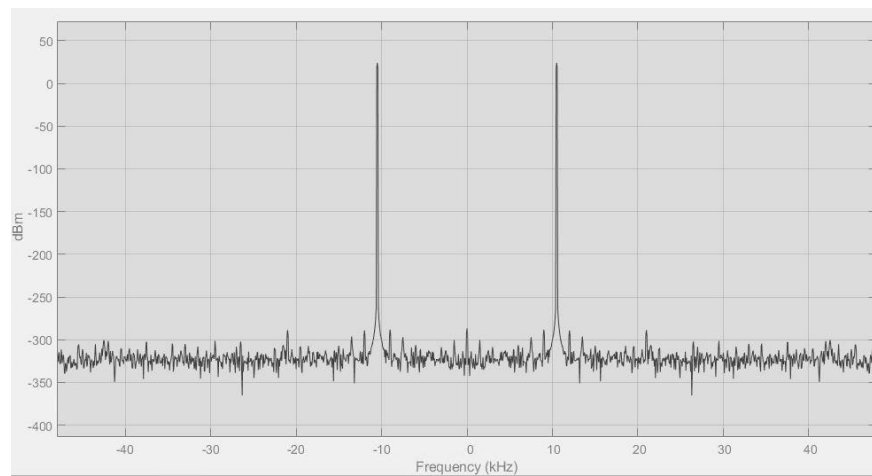


Figure 3. Spectrum Analyzer Display

### Discussion

- Why is the x-axis not in Hertz?
- Compare custom analyzer vs. built-in Spectrum Analyzer.
- Explore terms: dBm, negative frequency, noise floor.

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### Example 3: Simulating a Zero-Mean Square Wave

**Goal:** Generate a square wave with no DC component.

#### Blocks Used

- Pulse Generator (Sample-based, 50% duty cycle)
- Gain and Add (to remove DC)

- DSP Constant

### Setup

- Sampling Rate: 96 kHz
- For 1 kHz square wave: 96 samples/period, 48 samples pulse width

### Discussion

- Analyze spectrum: What frequencies are present?
- What happens if DC is not removed?
- Bandwidth implications for transmission.

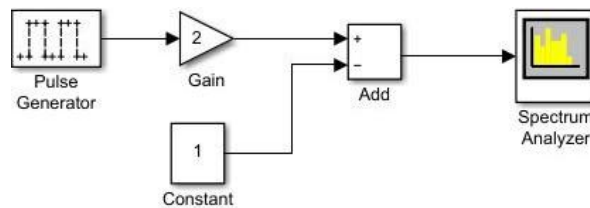


Figure 4. Zero-mean square wave generator

### Example 4: Multiplying Two Sinusoids

**Goal:** Understand basic Amplitude Modulation (AM).

### Setup

- Two DSP Sine Waves: 1.5 kHz and 12 kHz
- Product block to multiply
- Output to Spectrum Analyzer

### Discussion

- Observe sidebands in spectrum.
- Use trigonometric identity:  $\cos(a)\cos(b)$
- Introduce carrier frequency concept.

### Example 5: Synthesizing a Square Wave

**Goal:** Recreate a square wave by summing sinusoids.

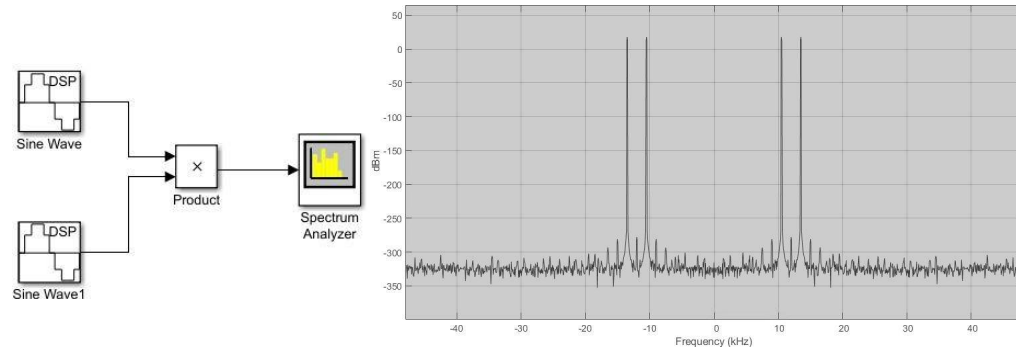
### Steps

- Use multiple DSP Sine Wave blocks.
- Set appropriate frequencies and amplitudes (based on Fourier Series).
- Sum using Add blocks.

- Visualize using Spectrum Analyzer (set to RMS mode).

### Discussion

- Compare synthesized wave to ideal square wave.
- Validate Fourier Series representation.
- Understand RMS and its relevance.



(a) Simulation Model

(b) Frequency Domain Display

Figure 5. Two Sine Waves Multiplied: Model and Spectrum



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Experiment No: 2 (Hardware)

Experiment Name: - Double Sideband Amplitude Modulation (AM) Generation

**Objectives:**

1. To investigate the generation of double sideband amplitude modulated (AM) waveforms.
2. To examine the effects of changing audio frequency and amplitude on carrier suppression.

**Equipment:**

- 2) IT-4101 Trainer Board
- 3) 2mm Patch Cords
- 4) Oscilloscope.

**Introduction:**

In telecommunications, **modulation** is the process of conveying a message signal, for example a digital bit stream or an analog audio signal, inside another signal that can be physically transmitted. Modulation of a sine waveform is used in view to transform a baseband message signal to a pass-band signal, for example a radio-frequency signal (RF signal). Modulating a sine wave carrier makes it possible to keep the frequency content of the transferred signal as close as possible to the Centrum frequency (typically the carrier frequency) of the pass-band.

**Analog Modulation Types:**

A low-frequency message signal (top) may be carried by an AM or FM radio wave. Common analog modulation techniques are:

- Amplitude modulation (AM) (here the amplitude of the carrier signal is varied in accordance to the instantaneous amplitude of the modulating signal)
  - Double-sideband modulation (DSB)
    - Double-sideband modulation with carrier (DSB-WC) (used on the AM radio broadcasting band)
    - Double-sideband suppressed-carrier transmission (DSB-SC)
  - Single-sideband modulation (SSB, or SSB-AM),
    - SSB with carrier (SSB-WC)
    - SSB suppressed carrier modulation (SSB-SC)

**Frequency Components of Human Voice**

When we speak, we generate a sound that is very complex and changes continuously so at a particular instant in time the waveform may appear as shown in Figure 1 below.

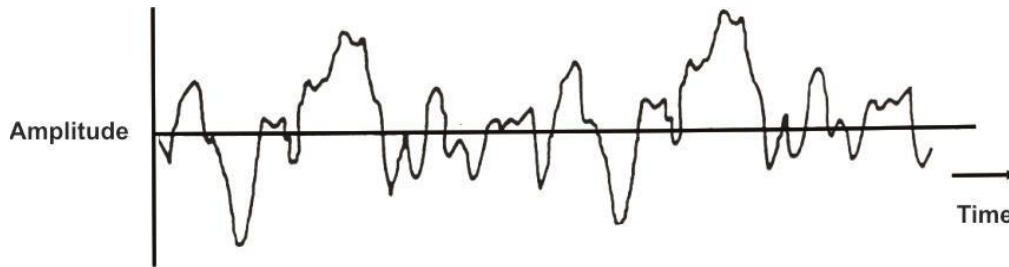


Fig. 1

### Amplitude Modulation (AM)

In amplitude modulation, the information signal is used to control the amplitude of the carrier wave. As the information signal increases in amplitude, the carrier wave is also made to increase in amplitude. Likewise, as the information signal decreases, then the carrier amplitude decreases.

By looking at Fig. 2 below, we can see that the modulated carrier wave does appear to 'contain' in some way the information as well as the carrier. To emphasize the connection between the information and the final waveform, a line is sometimes drawn to follow the peaks of the carrier wave as shown in Fig. 2. This shape, enclosed by a dashed line in our diagram, is referred to as an 'envelope', or a 'modulation envelope'.

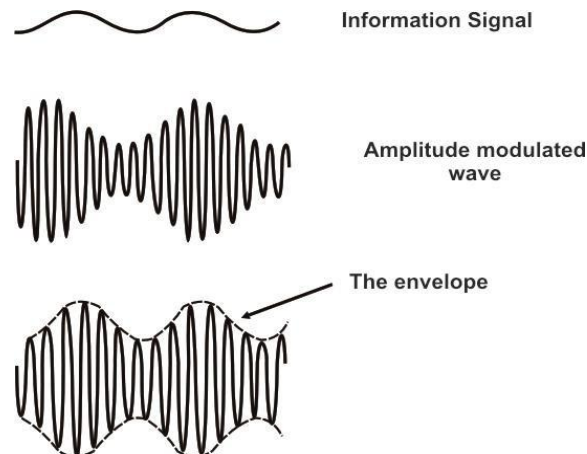


Fig. 2: AM signal

### Depth of Modulation

The amount by which the amplitude of the carrier wave increases and decreases depends on the amplitude of the information signal and is called the 'depth of modulation' or modulation index. The depth of modulation can be quoted as a fraction or as a percentage.

$$\text{Percentage modulation} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \times 100\%$$

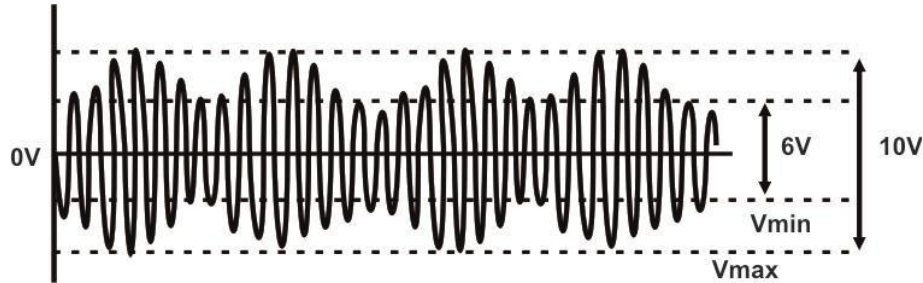


Fig. 3: Example to calculate modulation index

Assume a carrier frequency ( $f_c$ ) of 1 MHz and amplitude of, say 5 volts peak-to-peak. If we also have a 1 KHz information signal, or modulating frequency ( $f_m$ ), with amplitude of 2V peak-to-peak. When both signals have passed through the amplitude modulator they are combined to produce an amplitude modulated wave.

The resultant AM signal has a new frequency spectrum as shown in Fig. 4 inserting changes that occurs as a result of the modulation process:

1. The original 1 KHz information frequency has disappeared.
2. The 1 MHz carrier is still present and is unaltered.

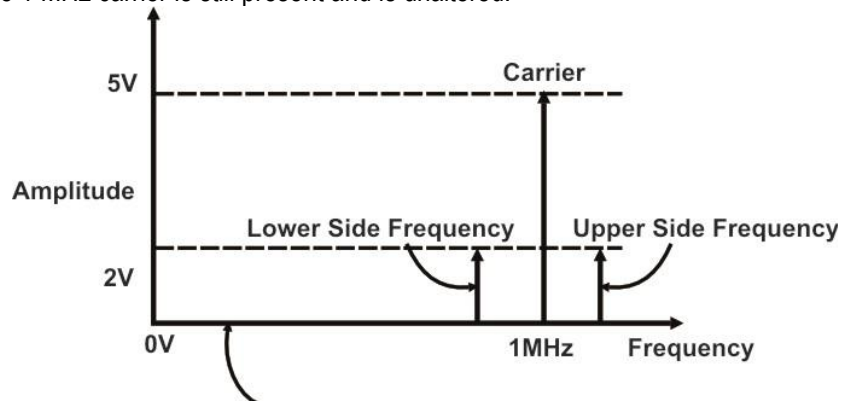


Fig. 4

There are two new components:

1. Carrier frequency ( $f_c$ ) plus the information frequency, called the upper side frequency ( $f_c + f_m$ ).
2. Carrier frequency ( $f_c$ ) minus the information frequency, called the lower side frequency ( $f_c - f_m$ ).

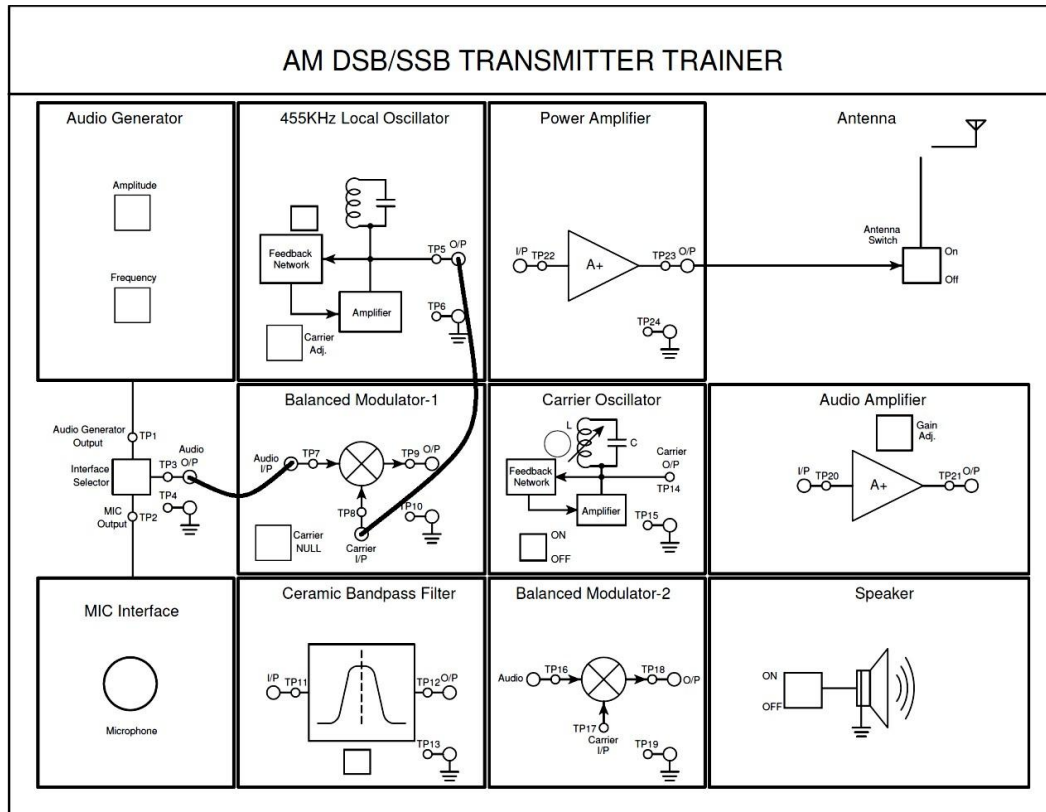
The resulting signal in this example has a maximum frequency of 1001 KHz and a minimum frequency of 999 KHz and so it occupies a range of 2 KHz. This is called the bandwidth of the signal. Notice how the bandwidth is twice the highest frequency contained in the information signal.

### Sidebands:

If the information signal consisted of range of frequencies, each separate frequency will create its own upper side frequency and lower side frequency.

(Note: Put the Carrier Oscillator switch in OFF position as this will distort the other outputs.)

## Experimental Setup:



## Procedure:

1. Turn on the power of IT-4101 module.
2. Make the select switch in Audio Generator Output position.
3. Connect Oscilloscope probe to Audio O/P test point TP3 and examine a low frequency waveform. Set the frequency to nearly 2 KHz. And set the amplitude of audio signal to maximum.
4. Connect 455KHz Carrier TP5 to Carrier I/P TP8 of Balanced Modulator. Carrier Null potentiometer should be adjusted to get a Carrier of maximum amplitude.
5. Now connect the Audio output TP3 to the Audio input TP7 of Balanced Modulator-1
6. Now connect the oscilloscope probe to "O/P" of Balanced Modulator-1 TP9, and observe the amplitude modulation.
7. Vary the Frequency and Amplitude of Audio generator and observe the changing in the modulated waveform as shown in Fig. 5.

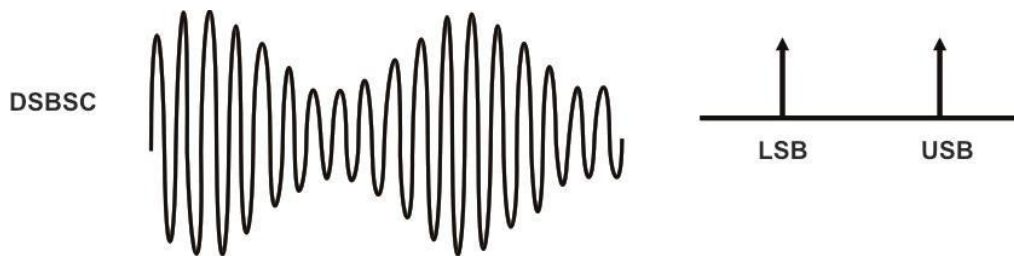


Fig. 5



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Experiment No: 3(a) (Hardware)

Experiment Name: - Single Sideband AM Generation

**Objective:**

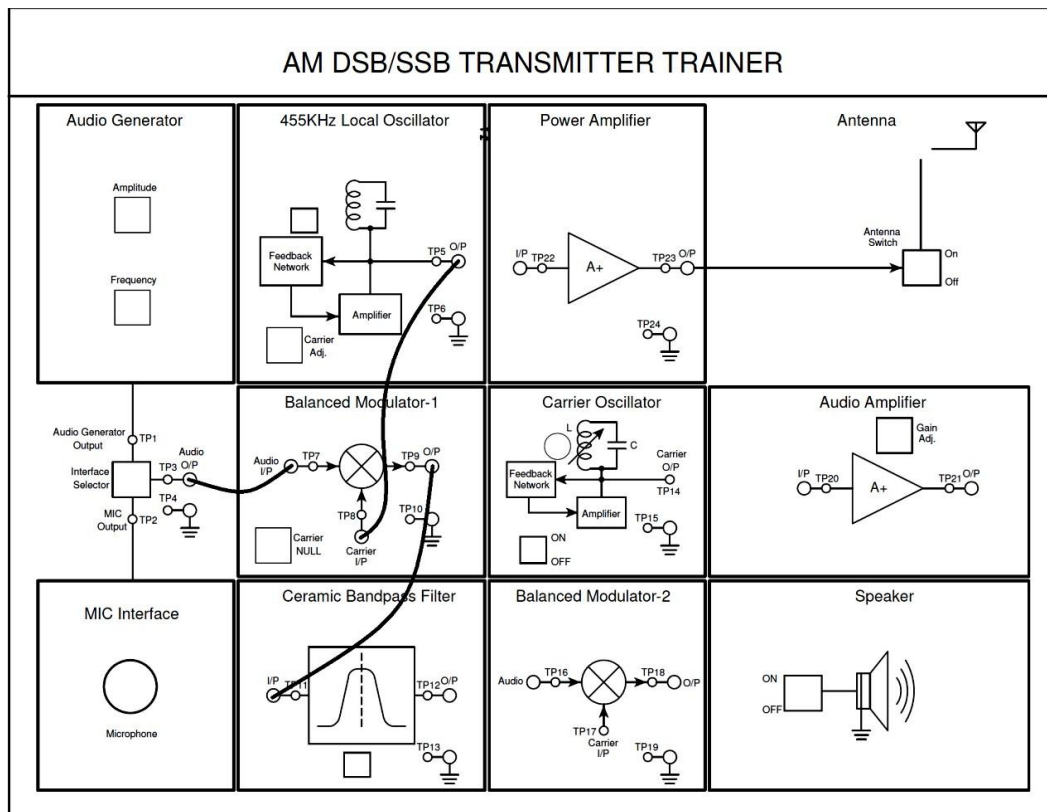
- 1) To investigate the generation of single sideband amplitude modulated (AM) waveforms.
- 2) To examine the effects of changing audio frequency and amplitude on carrier suppression.

**Equipment:**

- 1) IT-4101 Trainer Board
- 2) 2mm Patch Cords
- 3) CRO

**Experimental Setup**

Refer to the following diagram to configure setup for the present experiment



**Procedure:**

1. Turn on the power of IT-4101 module.

2. Make the select switch in Audio Generator Output position.
3. Connect Oscilloscope probe to Audio O/P test point TP3 and examine a low frequency waveform. Set the frequency to nearly 2 KHz. And set the amplitude of audio signal to maximum.
4. Connect 455KHz Carrier TP5 to Carrier I/P TP8 of Balanced Modulator. Carrier Null potentiometer should be adjusted to get a Carrier of minimum amplitude.
5. Now connect the Audio output TP3 to the Audio input TP7 of Balanced Modulator-1
6. Now connect the oscilloscope probe to "O/P" of Balanced Modulator-1 TP9, and observe the amplitude modulation.
7. Vary the Frequency and Amplitude of Audio Generator section and observe the changing in the modulated waveform as shown in Fig. 1.

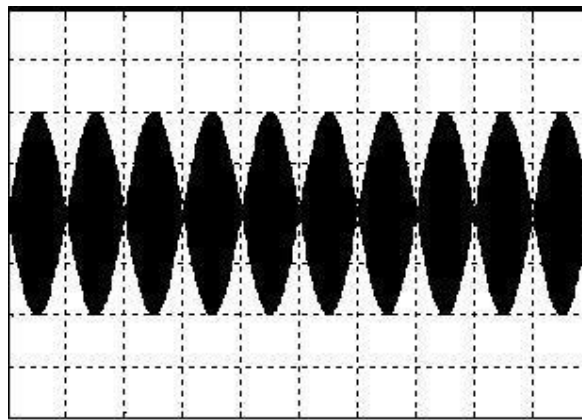


Fig. 1

8. Connect Balance Modulator O/P TP9 to Ceramic Filter I/P TP11 and observe the wave form at the output of Ceramic Filter TP12. The filter will completely remove the upper sideband of amplitude modulated signal. And observed signal will be SSB signal shown in Fig. 2.

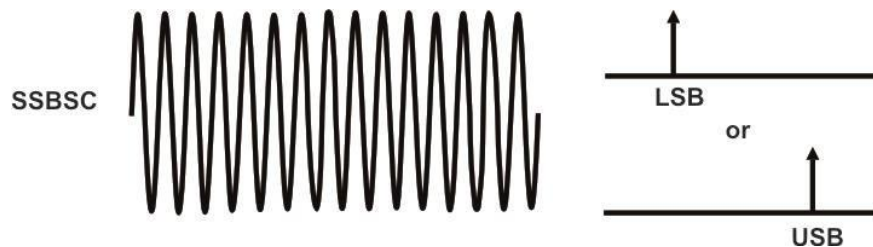


Fig. 2

**(Note: Put the Local Oscillator switch in OFF position as will distort the other outputs.)**



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Experiment No: 3(b) (Hardware)

Experiment Name:- **Demodulation of AM SSB-SC Modulated signal using Product Detector**

**Requirements**

1. IT-4101 Trainer Board
2. IT-4102 Trainer Board
3. 2mm Patch Cords
4. CRO

**Experimental Setup**

Refer to the following diagram to configure setup for the present experiment

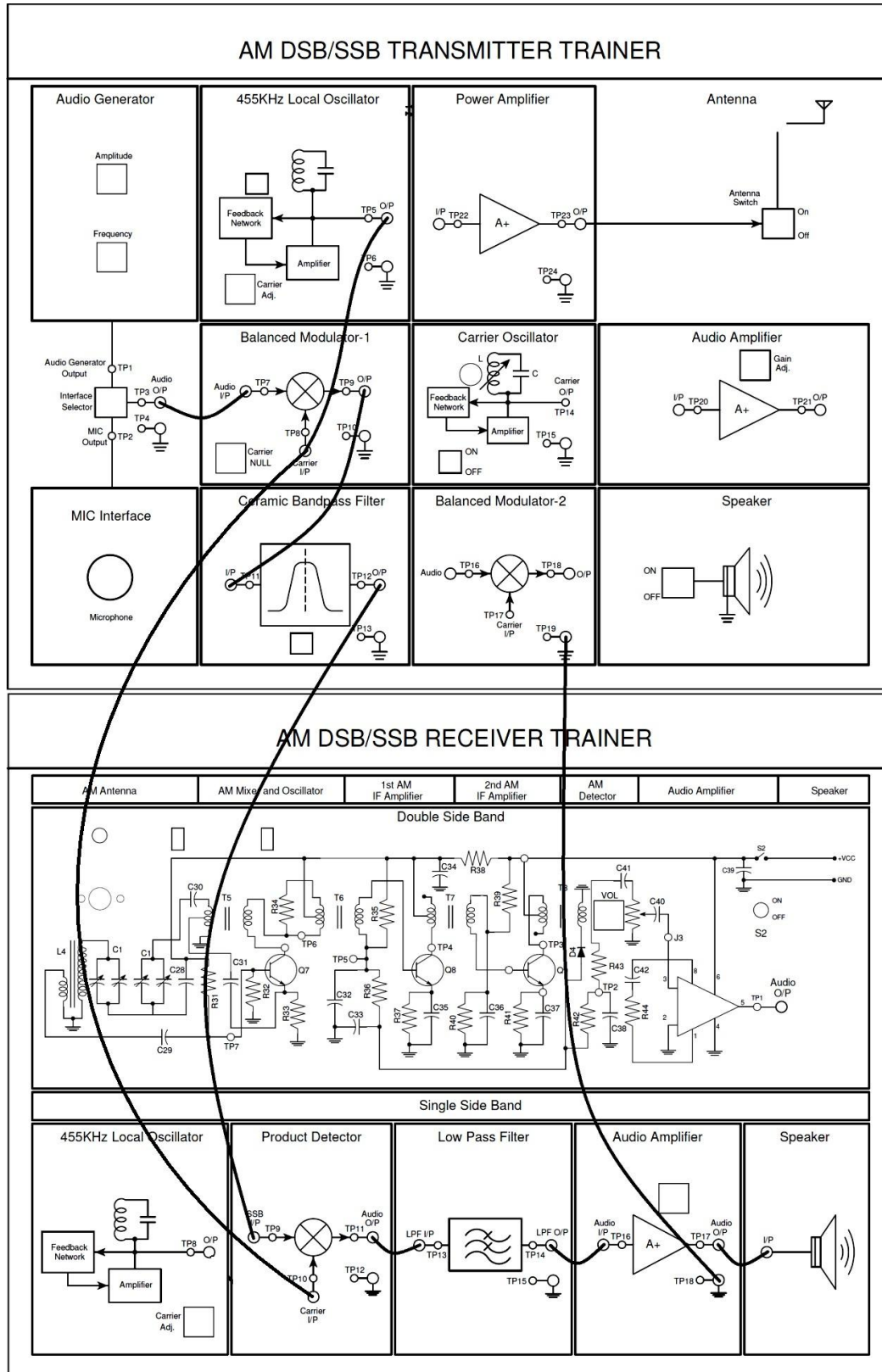


Fig 3.1

**Procedure:**

1. Turn on the power of IT-4101 module.
2. Make the select switch in Audio Generator Output position.
3. Connect Oscilloscope probe to Audio O/P test point TP3 and examine a low frequency waveform. Set the frequency to nearly 2 KHz. And set the amplitude of audio signal to maximum.
4. Connect 455KHz Carrier TP5 to Carrier I/P TP8 of Balanced Modulator. Carrier Null potentiometer should be adjusted to get a Carrier of minimum amplitude.
5. Now connect the Audio output TP3 to the Audio input TP7 of Balanced Modulator-1
6. Now connect the oscilloscope probe to "O/P" of Balanced Modulator-1 TP9, and observe the amplitude modulation.
7. Vary the Frequency and Amplitude of Audio Generator section and observe the changing in the modulated waveform as shown in Fig 3.2

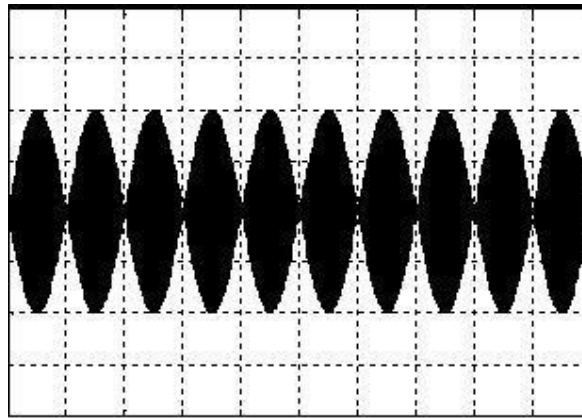


Fig 3.2

8. Connect Balance Modulator O/P TP9 to Ceramic Filter I/P TP11 and observe the wave form at the output of Ceramic Filter TP12. The filter will completely remove the upper sideband of amplitude modulated signal. And observed signal will be SSB signal shown in Fig 3.3

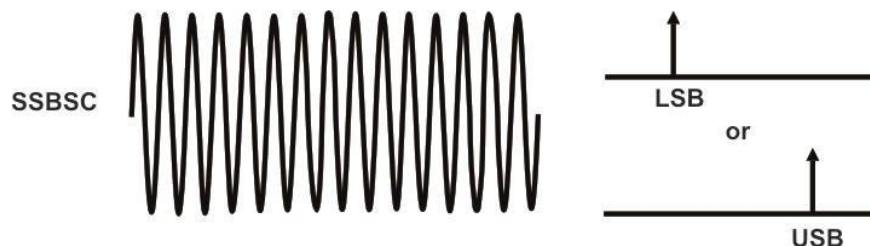


Fig 3.3

**(Note: Put the Local Oscillator switch in OFF position as will distort the other outputs.)**

9. Connect Ground of IT-4101 TP19 to Ground of IT-4102 TP18
10. Turn on the power of IT-4102 module
11. Connect output of Ceramic Band Pass Filter TP12 of IT-4101 to the input of Product Detector TP9 of IT-4102.
12. Connect 455 KHz Local Carrier of IT-4101 O/P TP5 to the Carrier Input TP10 of Product Detector of IT-4102.
13. Now observe the Audio O/P of Product Detector TP11 of module IT-4102. This will be noisy signal. Now vary the frequency of Audio generator of Module IT- 4101 and observe a corresponding change in frequency at Audio O/P of module IT-4102.
14. Connect the Audio O/P of Product Detector TP11 to Low Pass Filter TP13 of module IT-4102. This filter will eliminate the noise from the signal.
15. In order to amplify this signal connect O/P of LPF TP14 to I/P of Audio Amplifier TP16 of IT-4102 and connect O/P of Audio Amplifier TP17 to I/P of Speaker provided on IT-4102
16. Adjust the volume to hear the tone transmitted by IT-4101.



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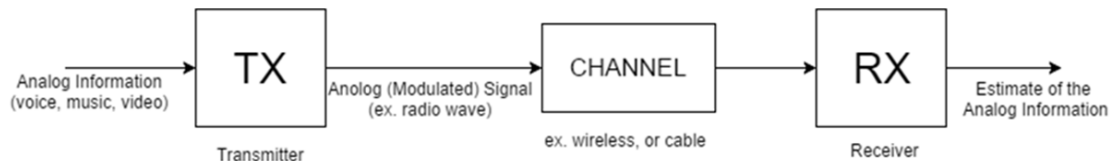
Experiment No: 2 (Software)

Experiment Name:- Double Sideband Amplitude Modulation & Demodulation

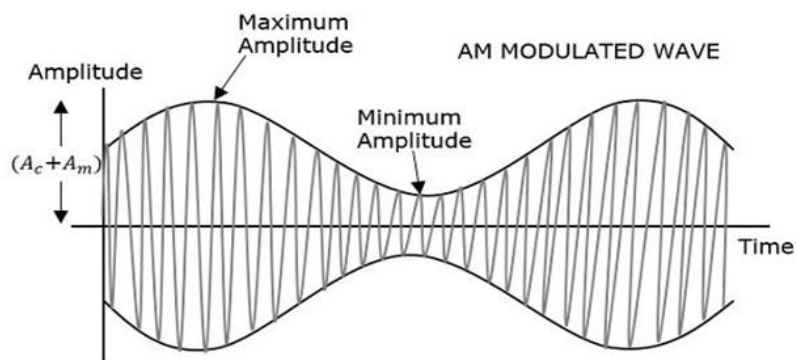
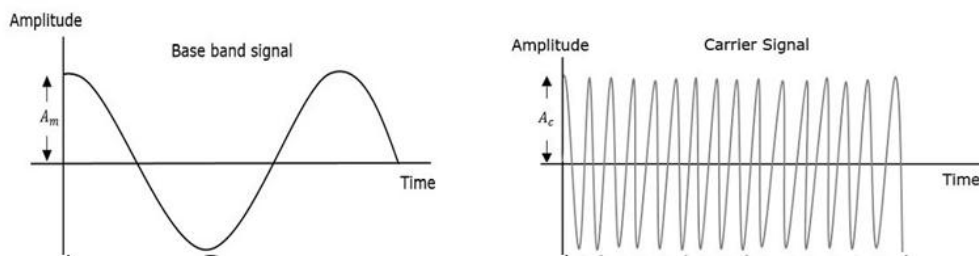
**Objective:** To perform the DSBAM and DSBSC signal Generation and Detection.

**Theoretical Background:**

**Fundamentals of Analog Communications:**



**Amplitude Modulation:**



## Mathematical Expressions:

### Time-domain Representation of the Waves:

Let, the modulating signal be,  $m(t)=A_m\cos(2\pi f_m t)$  and the carrier signal be,  $c(t)=A_c\cos(2\pi f_c t)$

Then, the equation of Amplitude Modulated wave will be

$$\begin{aligned}s(t) &= [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t) \\ s(t) &= A_c \left[ 1 + \left( \frac{A_m}{A_c} \right) \cos(2\pi f_m t) \right] \cos(2\pi f_c t) \\ \Rightarrow s(t) &= A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)\end{aligned}$$

Where,

- $A_m$  and  $A_c$  are the amplitude of the modulating signal and the carrier signal respectively.
- $f_m$  and  $f_c$  are the frequency of the modulating signal and the carrier signal respectively.

Here,  $\mu = \frac{A_m}{A_c}$

Where,  $\mu$  is Modulation index and it is equal to the ratio of  $A_m$  and  $A_c$ .

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

$A_{\max}$  and  $A_{\min}$  are the maximum and minimum amplitudes of the modulated wave.

### Problem Statement:

- Draw all the models one-by-one in the *MATLAB Simulink* window.
- Set the simulation time from the model window as below:

*SimulationàSimulation parameters: Start time = 0.0 & Stop time = 10.0.*

- Then simulate all the models and observe all the wave shapes at different points of the model.

### Software and Device requirement:

MATLAB (with simulink)

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit);

Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012

Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU  
Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

### Procedure

- You can take the blocks from the *Simulink Library Browser*.

### Blocks to be used for this experiment:

- The blocks that you will use in this experiment can be taken from the following mentioned library:

Bernoulli Binary Generator: Communication Blockset→Comm sources→Data Sources→Bernoulli Binary Generator

Sine Wave: Simulink Sources→Sine wave

Product: Simulink→Math operations→Product

AWGN Channel: Communication Block Set→Channels→AWGN Channel

Digital Filter Design: Communication Blockset→Comm Filters→Filter Designs Library Link→Digital Filter Design

Analog/Digital Filter Design: Signal Processing Blockset→Filtering→Filter Implementations→Analog/Digital Filter Design

Spectrum Scope: Signal Processing Blockset →Signal Processing Sinks→Spectrum Scope

Compare to Constant: Simulink→Logic & Bit Operations→Compare to Constant

Scope: Simulink→Sinks→Scope

Constant: Simulinks→Sources→Constants

Abs: Simulink→Math operations→Abs

Sum: Simulink→Math Operations→Sum

### Parameters to be used for different blocks:

- Spectrum Scope:
  - Check 'Show Scope Properties' & 'Buffer input'. Set: Buffer size: 128, Buffer overlap: 64 & Number of spectral averages: 2.
  - Check 'Show axis properties'. Set: Frequency unit: Hertz & Frequency range:  $[-Fs/2 \dots Fs/2]$ .
  - After Simulation you will observe a window for each spectrum scope. To view the spectrum more clearly, set Axes to Autoscale from the spectrum window.
- Sine wave:
  - Both modulator & demodulator:  
  
Sine type: Time based Amplitude: 1.0  
  
Frequency:  $2\pi \cdot 1$  rad/sec  
  
Sample time: 1/1000
- For carrier signal: Sine type: Time based Amplitude: 2.0 Frequency:  $2\pi \cdot 20$  rad/sec Sample time: 1/1000
- Product: Number of inputs: 2 & Multiplication: Element wise
- AWGN Channel: Initial seed: 67, Mode: Variance from mask & variance: 1.

Note: You can vary the noise power by varying variance

- Digital Filter Design:  
  
Filter type: Lowpass  
  
Design method: FIR to Equiripple  
  
Filter order: Minimum order  
  
Frequency Specifications: Units: Hz, FS: 5000, Fpass=5, Fstop=8.  
  
Magnitude Specifications: Units: dB, Apass=1, Astop=80.

Then click Design Filter. The magnitude response of the filter will be changed according to your specifications.

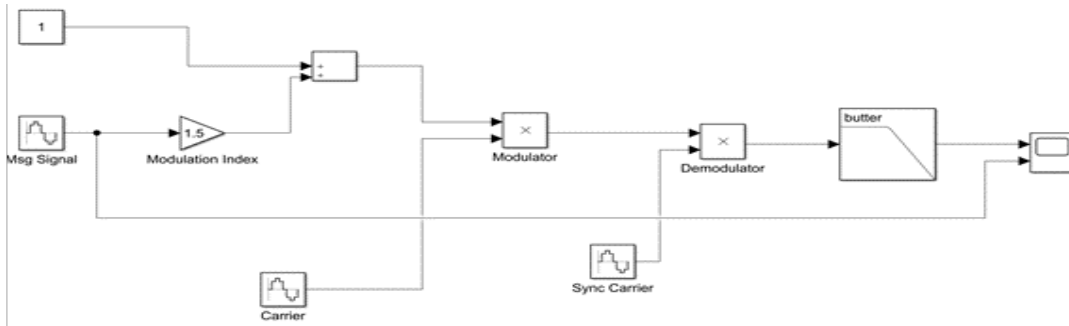
Note: You should choose your Fpass and Fstop frequencies depending on the frequency of your modulating signal.

- Compare to constant: Operator:  $\geq$ . Constant: Depends on your system. This is the threshold value of the detector.

- Scope: *Data History: Uncheck the 'Limit data points to last'.*
- Constant: *Constant value: 1*
- Abs: *Check the 'saturate on integer overflow' & 'Enable zero crossing detection'.*
- Sum: *List of signs: |+- (subtraction will be done).*

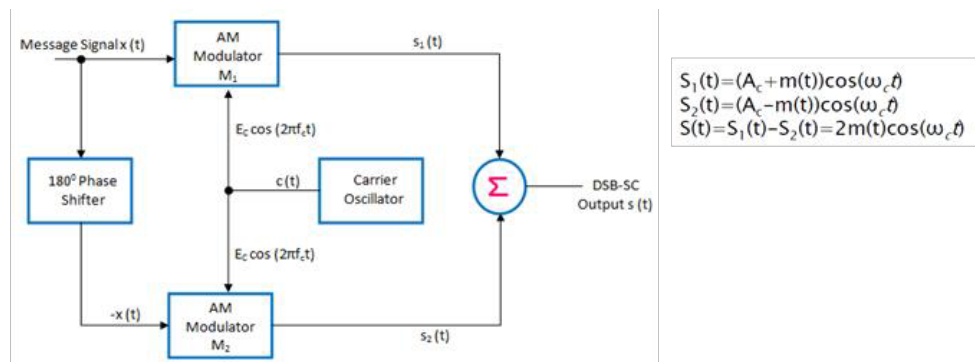
Note: To sum up two inputs, list of signs: |++

### Simulation Model



### Report:

- Add AWGN channel (to see the effect of noise) in between your modulator and demodulator and observe the output. Include your model diagram and output waveforms (choose different modulation index value for under, over & perfect modulation) in the report.
- Build the following block of Balanced Modulator to generate DSBSC waveform in MATLAB Simulink. Observe the modulated output. Also demodulate (Synchronous Demodulation) the DSBSC output to extract the message signal.





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**EEE/ECE341L/342: Introduction to Communication Engineering**  
**Laboratory**

Experiment No: 3 (Software)

Experiment Name:- Amplitude Modulation (AM) Single Sideband Suppressed Carrier (SSB-SC)- Modulation and Demodulation

**Objective:** To perform the AM SSB-SC signal Generation and Detection.

**Theoretical Background:**

**Advantages of SSB-SC AM:**

- DSB wastes Bandwidth: If bandwidth of message is B (in Hz), then bandwidth of  $\psi_{DSB}$  signal is 2B (in Hz) and Bandwidth is an important constraint in today's congested spectrum

Solution: Use  $\psi_{SSB}$  signal, that has bandwidth B in Hz

SSBSC: One sideband  $\rightarrow$  BW reduction

- DSBSC: No carrier  $\rightarrow$  reduction of power by 50%
- SSBSC: Only one sideband  $\rightarrow$  reduction of power by 50% more

**Mathematical Expressions:**

Let, the modulating signal be,  $m(t)$  and the carrier signal be  $\cos \omega_c(t)$

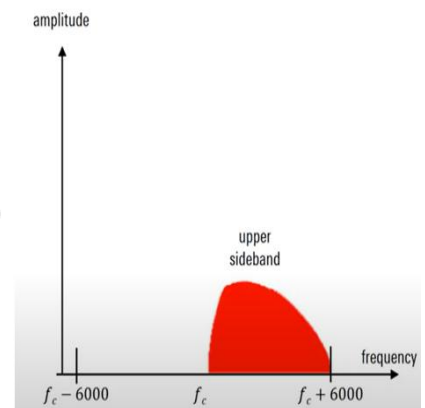
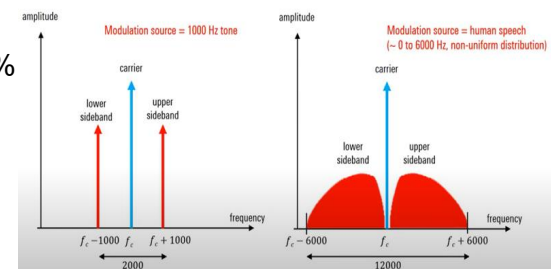
SSB modulated output:  $\psi_{SSB}(t) = m(t) \cdot \cos(\omega_c t) \mp m_h(t) \cdot \sin(\omega_c t)$

where  $m_h(t)$  = Hilbert Transform of  $m(t)$  =  $-\pi/2$  phase-shifted version of  $m(t)$

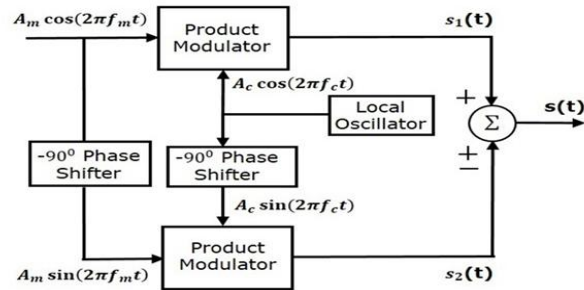
- Multiply by carrier for demodulation:

$$\begin{aligned} \psi_{SSB}(t) \cdot \cos(\omega_c t) &= [m(t) \cdot \cos(\omega_c t) \mp m_h(t) \cdot \sin(\omega_c t)] \cdot \cos(\omega_c t) \\ &= \frac{1}{2} m(t) + \frac{1}{2} [m(t) \cdot \cos(2\omega_c t) \mp m_h(t) \cdot \sin(2\omega_c t)] \end{aligned}$$

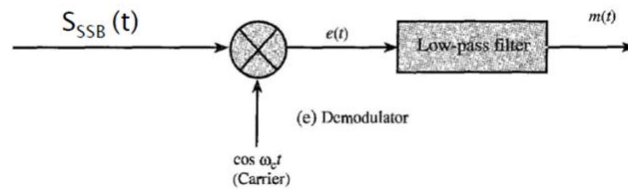
- Then use LPF to extract message signal



## SSB Modulation-



## SSB Demodulation-



## Problem Statement:

- Draw all the models one-by-one in the MATLAB Simulink window.
- Set the simulation time from the model window as below:  
*Simulation---->Simulation parameters: Start time = 0.0 & Stop time = 10.0.*
- Then simulate all the models and observe all the wave shapes at different points of the model.

## Software and Device requirement:

MATLAB (with simulink)

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit);

Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012

Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

**Procedure:**

You can take the blocks from the Simulink Library Browser.

The blocks that you will use in this experiment can be taken from the following mentioned library:

- (i) Bernoulli Binary Generator: *Communication Blockset--> Comm sources--> Data Sources--> Bernoulli Binary Generator*
- (ii) Sine Wave: *Simulink--> Sources--> Sine wave*
- (iii) Product: *Simulink--> Math operations--> Product*
- (iv) AWGN Channel: *Communication Block Set--> Channels--> AWGN Channel*
- (v) Digital Filter Design: *Communication Blockset--> Comm Filters--> Filter Designs Library Link--> Digital Filter Design*
- (vi) Analog/Digital Filter Design: *Signal Processing Blockset--> Filtering--> Filter Implementations -> Analog/Digital Filter Design*
- (vii) Spectrum Scope: *Signal Processing Blockset--> Signal Processing Sinks--> Spectrum Scope*
- (viii) Compare to Constant: *Simulink--> Logic & Bit Operations--> Compare to Constant*
- (ix) Scope: *Simulink--> Sinks--> Scope*
- (x) Constant: *Simulinks--> Sources--> Constants*
- (xi) Abs: *Simulink--> Math operations--> Abs*
- (xii) Sum: *Simulink--> Math Operations--> Sum*

**Parameters to be used for different blocks:****Spectrum Scope:**

- Check 'Show Scope Properties' & 'Buffer input'. Set: Buffer size: 128, Buffer overlap: 64 & Number of spectral averages: 2.
- Check 'Show axis properties'. Set: Frequency unit: Hertz & Frequency range:  $[-F_s/2 \dots F_s/2]$ .
- After Simulation you will observe a window for each spectrum scope. To view the spectrum more clearly, set Axes-->Autoscale from the spectrum window.

**Sine wave:**

Both modulator & demodulator:

Sine type: Time based

Amplitude: 1.0

Frequency:  $2\pi \cdot 1$  rad/sec

Sample time: 1/1000

For carrier signal:

Sine type: Time based

Amplitude: 2.0

Frequency:  $2\pi \cdot 20$  rad/sec

Sample time: 1/1000

**Product:** Number of inputs: 2 & Multiplication: Element wise

**AWGN Channel:** Initial seed: 67, Mode: Variance from mask & variance: 1.

### **Digital Filter Design:**

Filter type: Lowpass

Design method: FIR-->Equiripple

Filter order: Minimum order

Frequency Specifications: Units: Hz, FS: 5000, Fpass=5, Fstop=8.

Magnitude Specifications: Units: dB, Apass=1, Astop=80.

Then click Design Filter. The magnitude response of the filter will be changed according to your specifications.

Note: You should choose your Fpass and Fstop frequencies depending on the frequency of your modulating signal.

**Compare to constant:** Operator:  $\geq$ . Constant: Depends on your system. This is the threshold value of the detector.

**Scope:** Data History: Uncheck the 'Limit data points to last.'

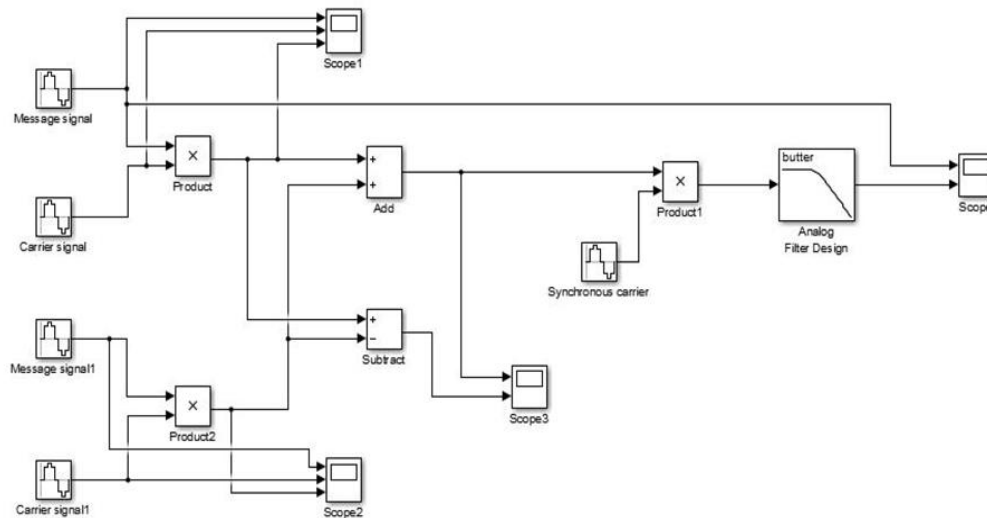
**Constant:** Constant value: 1

**Abs:** Check the 'saturate on integer overflow' & 'Enable zero crossing detection.'

**Sum:** List of signs:  $|+-$  (subtraction will be done).

Note: To sum up two inputs, list of signs:  $|++$

### Simulation Model:



### Lab Report:

1. Add AWGN channel (to see the effect of noise) in between your modulator and demodulator and observe the output. Include your model diagram and output waveforms (USB, LSB and demodulated waveforms) in the report.
2. Use the built in SSBAM modulator and demodulator from the library and do the simulation. Include your model diagram and output waveforms (Modulated and demodulated waveforms) in the report.

#### Additional Task: (Optional)

Use a spectrum analyzer from the library and show the USB, LSB and extracted output in frequency domain.

[Hint: Spectrum Analyzer settings: Type: RMS, Trace option-->Scale-->Linear]



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Experiment no: 4 (Hardware)

Experiment Name: - Frequency modulation and demodulation

**Objective:**

- 1) Demonstration of frequency modulation by Varactor modulator.
- 2) Demonstration of frequency demodulation by Foster-Seeley detector and effect of noise on its performance.

**Equipment:**

- 1) IT-4103 Trainer Board
- 2) 2mm Patch Cords
- 3) CRO

**Theoretical Background:**

Frequency Modulation (Fm)

**Frequency Modulation (FM)** conveys information over a carrier wave by varying its frequency (contrast this with amplitude modulation, in which the amplitude of the carrier is varied while its frequency remains constant). In analog applications, the instantaneous frequency of the carrier is directly proportional to the instantaneous value of the input signal. Digital data can be sent by shifting the carrier's frequency among a set of discrete values, a technique known as frequency-shift keying.

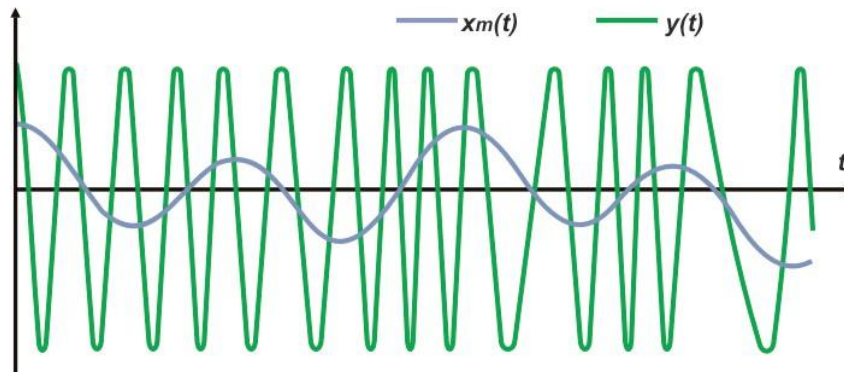


Figure 1: The modulated signal,  $y(t)$ , produced from frequency-modulating  $x_c(t)$  with  $x_m(t)$ .

The frequency of the carrier is made to increase as the voltage in the information signal increases and to decrease in frequency as it reduces. The larger the amplitude of the information signal, the further the frequency of the carrier signal is shifted from its starting point. The frequency of the information signal determines how many times a second this change in frequency occurs. Notice in Fig. 1 that the modulation process does not affect the amplitude.

## FM Transmitter

The block diagram of a typical FM Transmitter is shown in Fig. 2 below.

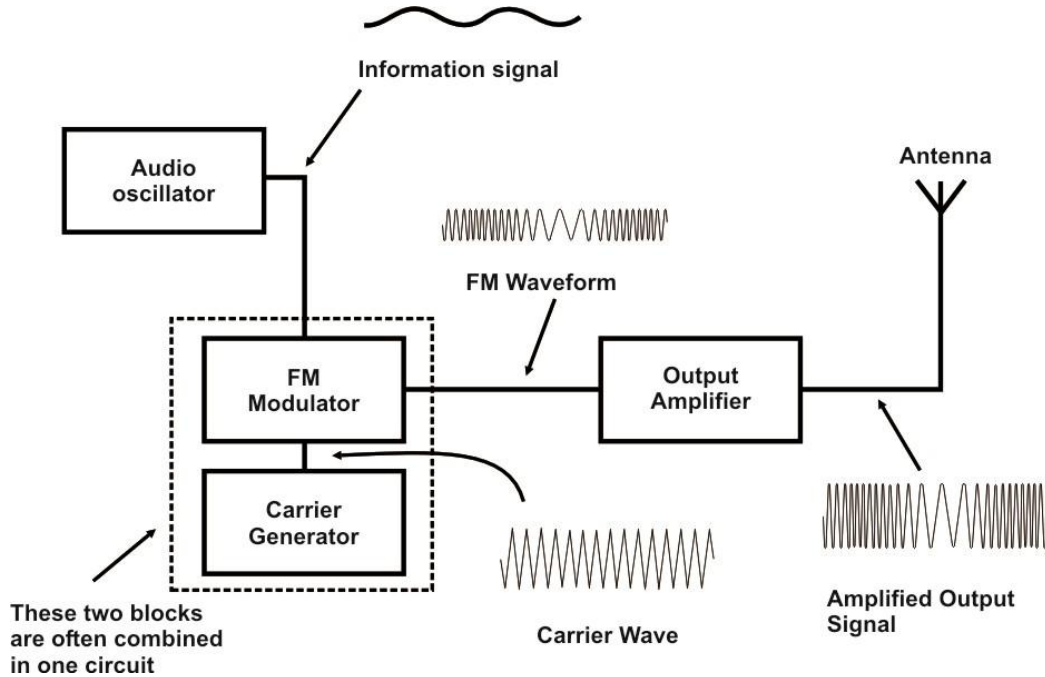


Fig. 2: Block diagram of FM transmitter

The audio oscillator supplies the information signal and can be replaced by a microphone and AF amplifier to provide speech and music instead of single sine wave frequency provided on IT-4103. There are various modulator, namely Varactor Modulator and the Reactance Modulator. Here, we will study only Varactor Modulator.

### Working Principle of Varactor Modulator

The basic idea is quite simple and both modulations function in much the same way. They both include an RF oscillator to generate the carrier and these oscillators employ a parallel tuned circuit to determine the frequency of operation. Adding an additional capacitor in parallel will cause the total capacitance to increase and this will result in a decrease in the resonance frequency.

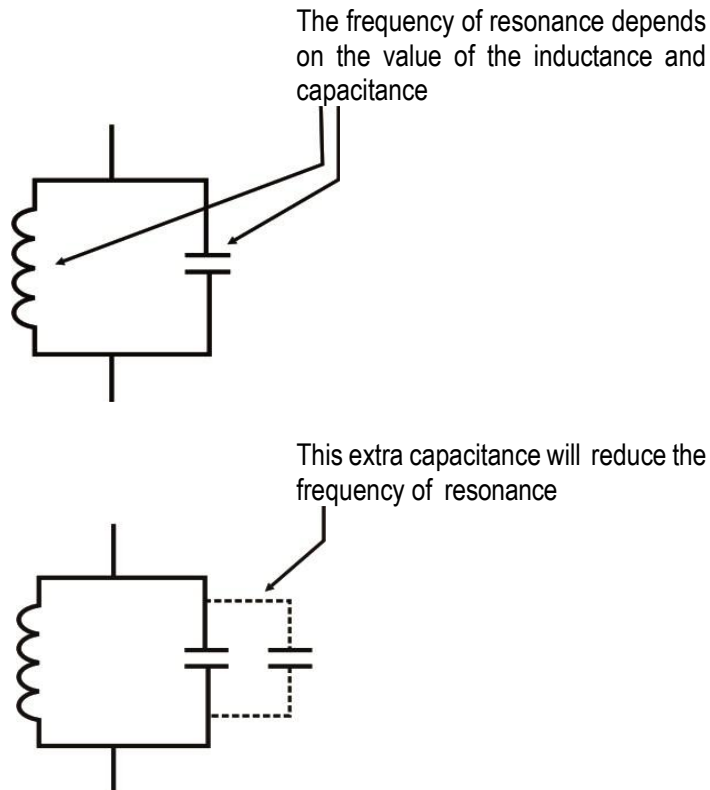


Fig. 3: Varactor Modulator

### Varactor Diode:

The varactor diode is a semiconductor diode that is designed to behave as a voltage controlled capacitor. When a semiconductor diode is reverse biased, no current flows and it consists of two conducting regions separated by non-conducting region. This is very similar to the construction of a capacitor.

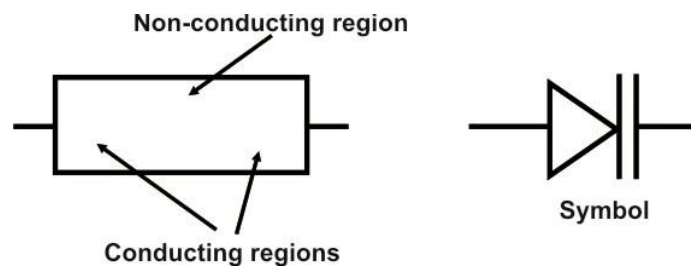


Fig. 4: Varactor Diode

By increasing the reverse biased voltage, the width of the insulating region can be increased and hence the capacitance value decreases. This is shown in Fig. 4.

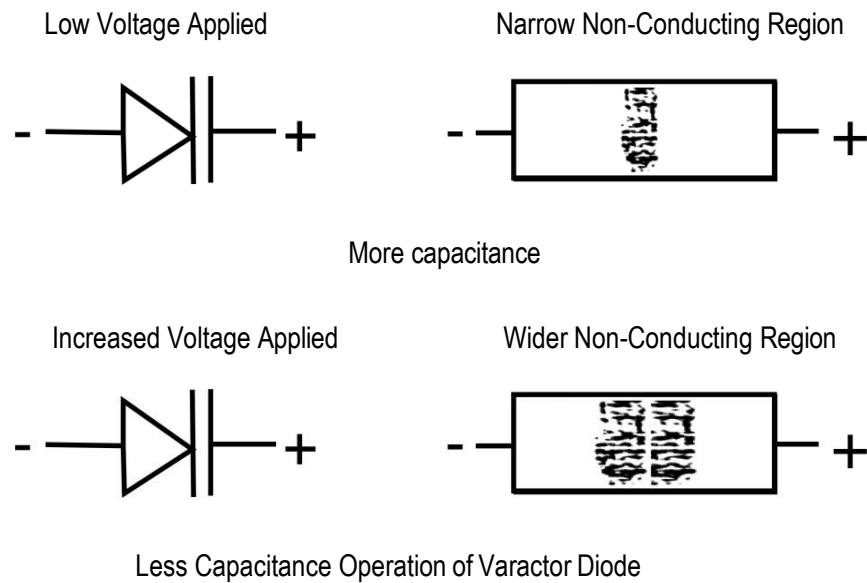


Fig. 4

If the information signal is applied to the varactor diode, the capacitance will therefore be increased and decreased in accordance with the incoming signal.

### Working Principle of a Reactance Modulator

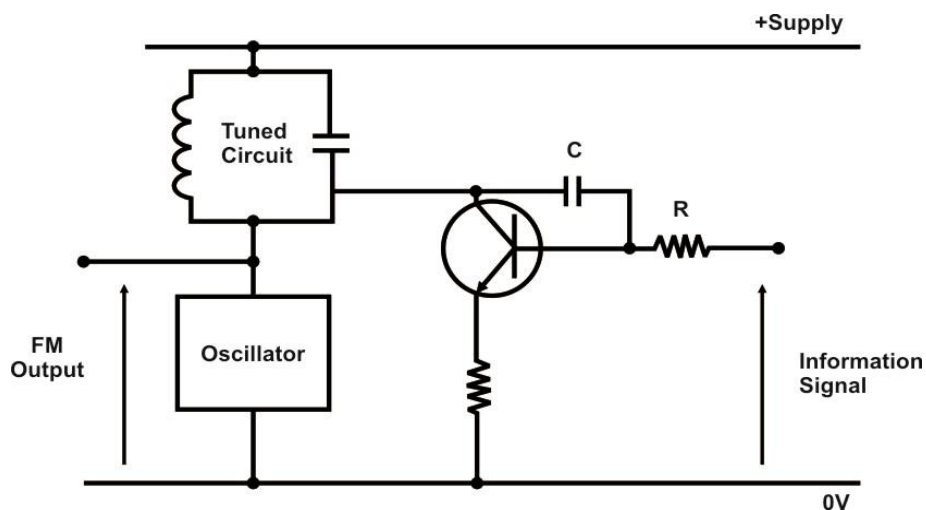


Fig. 5: Reactance modulator

In Fig. 5, the left hand half is the previous varactor modulator simply an oscillator and a tuned circuit, which generates the un-modulated carrier. The capacitor C and the resistor R are the two components used for the phase shifting, and together with the transistor, form the voltage controlled capacitor. This voltage-controlled capacitor is actually in parallel with the tuned circuit.

Figure 6 is more elaborate and helps in understanding the concept. In the first part of the Figure, the capacitor and associated components have been replaced by the variable capacitor, shown dotted.

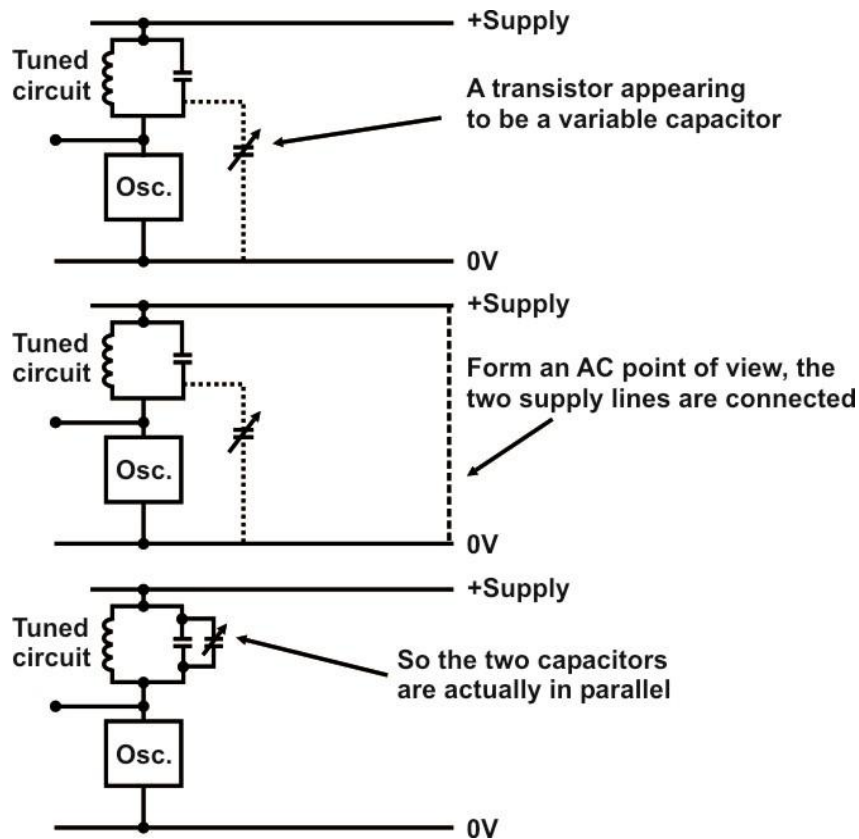


Fig.6

In the next part, the two supply lines are connected together. We can justify this by saying that the output of the DC power supply always includes a large smoothing capacitor to keep the DC voltages at a steady value.

This large capacitor will have a very low reactance at the frequencies being used in the circuit less than a milliohm. We can safely ignore this and so the two supply lines can be assumed to be joined together. Remember that this does not affect the DC potentials, which remain at the normal supply voltages.

If the two supply voltages are at the same AC potential, the actual points of connection do not matter and so we can redraw the circuit as shown in the third part.

Operation of the Reactance Modulator:

If required, reference can be made to Fig.5.

1. The oscillator and tuned circuit provide the un-modulated carrier frequency and this frequency is present on the collector of the transistor.
2. The capacitor and the resistor provide the  $90^\circ$  phase shift between the collector voltage and current. This makes the circuit appear as a capacitor.

- The changing information signal being applied to the base has the same effect as changing the bias voltage applied to the transistor and, this would have the effect of increasing and decreasing the value of this capacitance.

As the capacitance is effectively in parallel with the tuned circuit the variations in value will cause the frequency of resonance to change and hence the carrier frequency will be varied in sympathy with the information signal input.

### Experimental Setup (Modulation-1):

#### Part A-1: Frequency modulation by Varactor modulator.

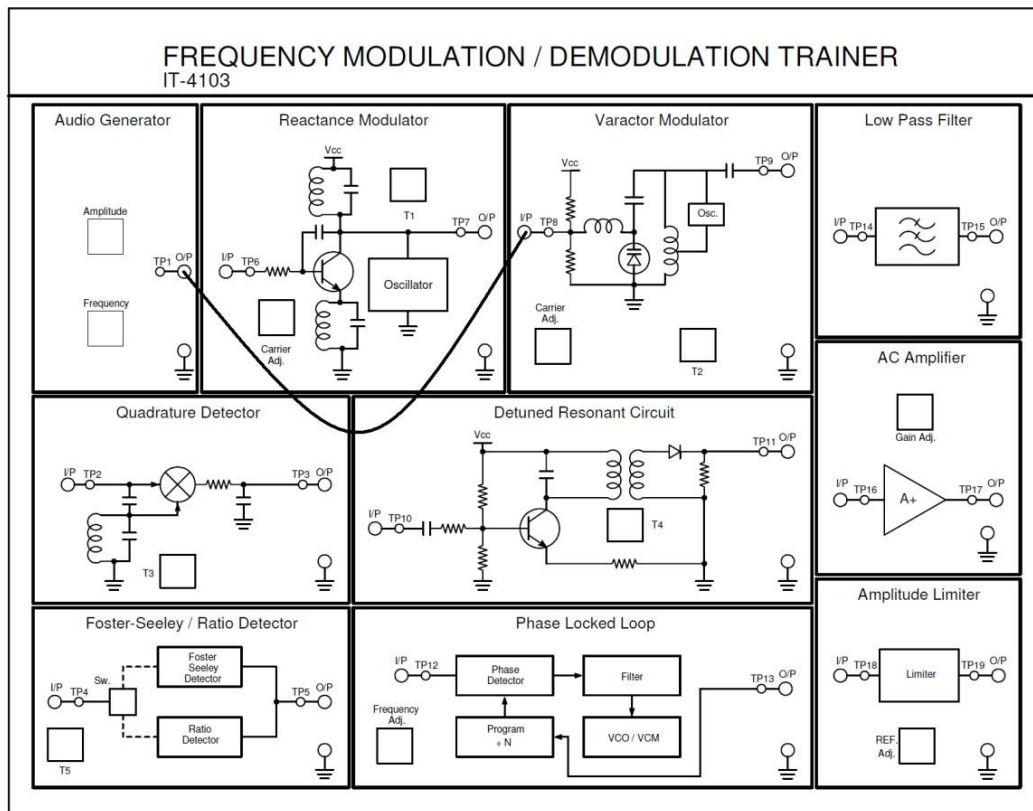


Fig. 7: Experimental Setup using Varactor modulator

#### Procedure

Upon completion of this experiment students will be able to understand the concept of Varactor Modulator.

- Connect O/P of Audio Generator TP1 to Ch 1 of Oscilloscope. Check the O/P by adjusting Amplitude and Frequency knob to 2 KHz.
- Connect O/P of Varactor Modulator TP9 to Ch 2 of Oscilloscope. Observe the signal at O/P and calculate the frequency of Carrier generated by this section. It should around 420 KHz. If not so, then using a tuner, turn the screw of the Coil gently while observing the O/P at Oscilloscope.
- Make the connections on Module IT-4103 as shown in Fig. 7 above.

4. Connect O/P of Varactor Modulator TP9 to Ch 2 of Oscilloscope. Now vary the amplitude of the Audio signal. Variation in frequency as shown in Fig. 8 can be observed.

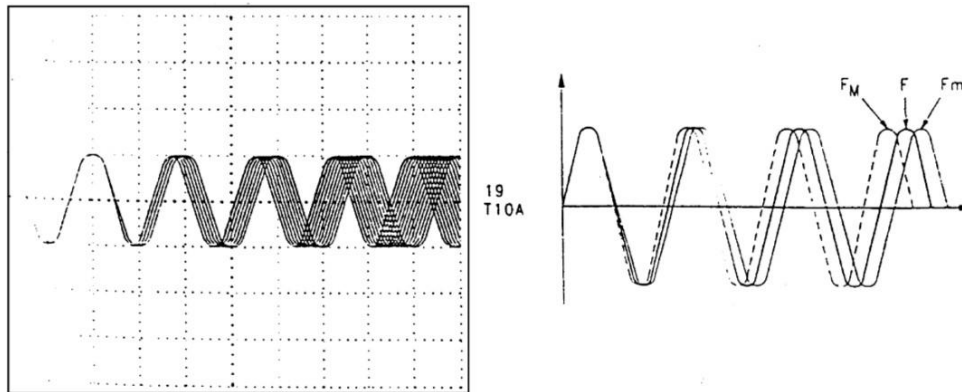


Fig. 8

### Experimental Setup (Modulation-2):

**Part A-2:** Frequency modulation by Reactance modulator.

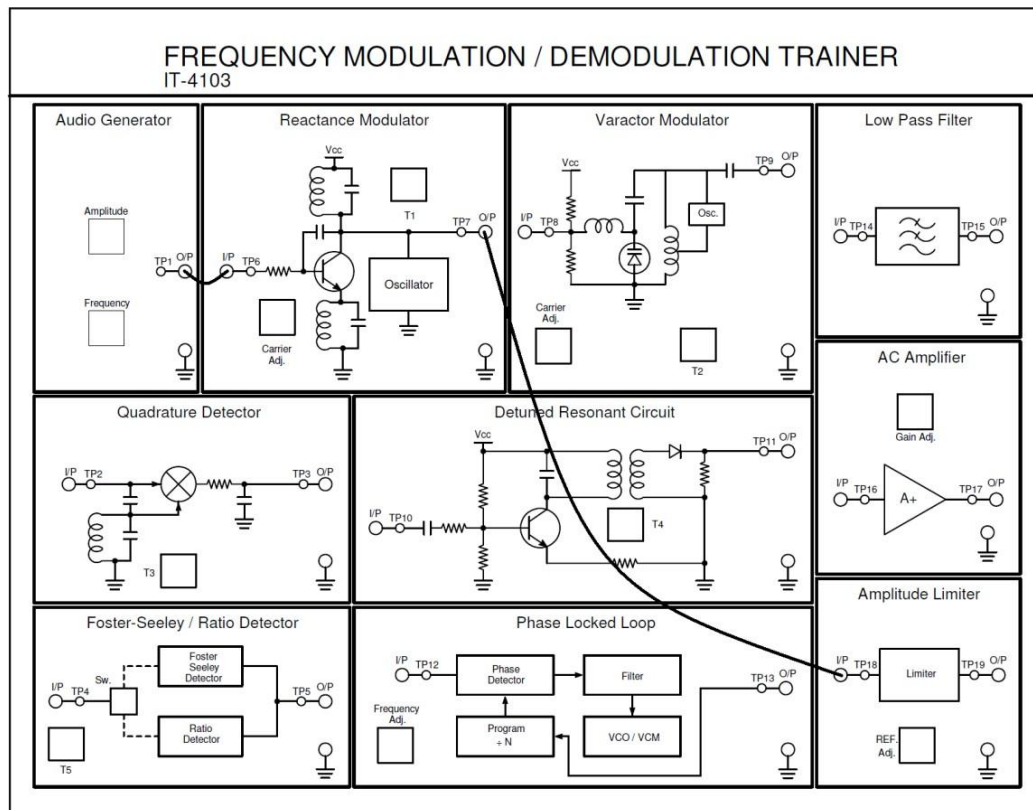
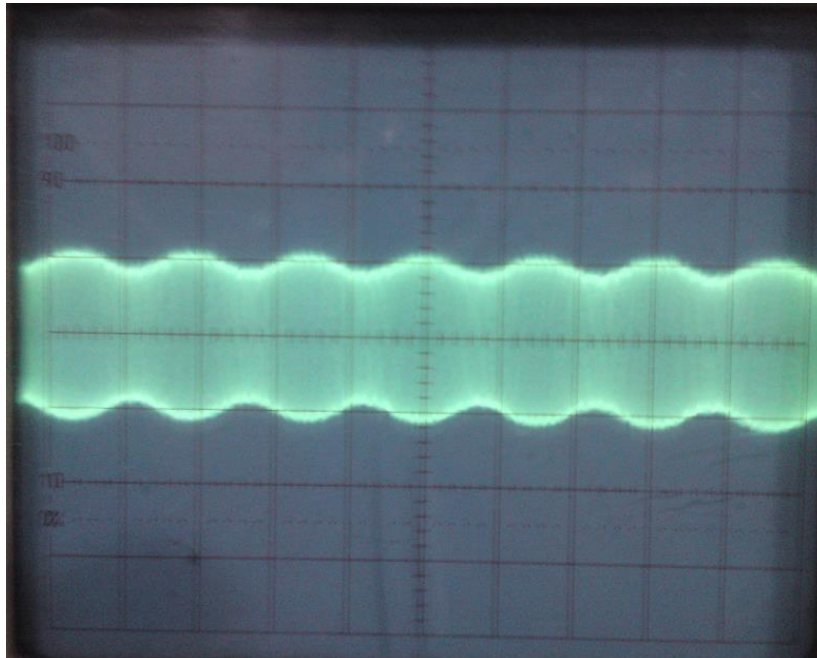


Fig. 9: Experimental Setup using Reactance modulator

## Procedure

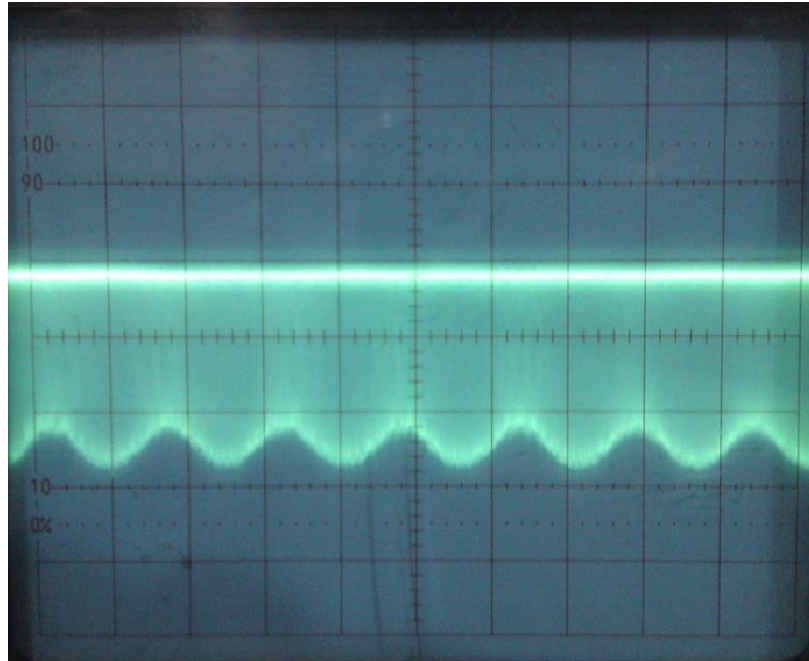
Upon completion of this experiment students will be able to understand the concept of Reactance Modulator.

1. Connect O/P of Audio Generator TP1 to Ch 1 of Oscilloscope. Check the O/P by adjusting Amplitude and Frequency knob to 2KHz.
2. Connect O/P of Reactance Modulator TP7 section to Ch 2 of Oscilloscope. Observe the signal at O/P and calculate the frequency of Carrier generated by this section. It should be around 370-390 KHz. If not so, then using a plastic tuner, turn the screw of the Coil gently while observing the O/P at Oscilloscope.
3. Make the connections on Module IT-4103 as shown in Fig. 9 above.
4. Connect O/P of Reactance Modulator TP7 section to Ch 2 of Oscilloscope.
5. Frequency Modulation O/P will be similar to that of Varactor Modulator with the exception that it will also contain AM part. This can be observed when we set the Time base of Oscilloscope to ms scale as shown in Figure below.



**Note:** there will be an Amplitude Modulated part added in the Frequency Modulated wave when using Reactance Modulator. To nullify it, Amplitude Limiter is used.

6. Connect output of Reactance Modulator TP7 to the input of Amplitude Limiter TP18.
7. Observe the output at TP19. The following wave must be formed on your CRO after passing through Amplitude Limiter. Now vary the amplitude and the frequency of the Audio signal. Variation in frequency can be observed.



**Note:** Vary the potentiometer of the Amplitude Limiter section so that the above waveform could be obtained.

### Operation of Foster - Seeley Detector

It uses a double-tuned RF transformer to convert frequency variations in the received FM signal to amplitude variations. These amplitude variations are then rectified and filtered to provide a dc output voltage. This voltage varies in both amplitude and polarity as the input signal varies in frequency. A typical discriminator response curve is shown in Fig. 10. The output voltage is 0 when the input frequency is equal to the carrier frequency ( $f_r$ ). When the input frequency rises above the center frequency, the output increases in the positive direction. When the input frequency drops below the center frequency, the output increases in the negative direction.

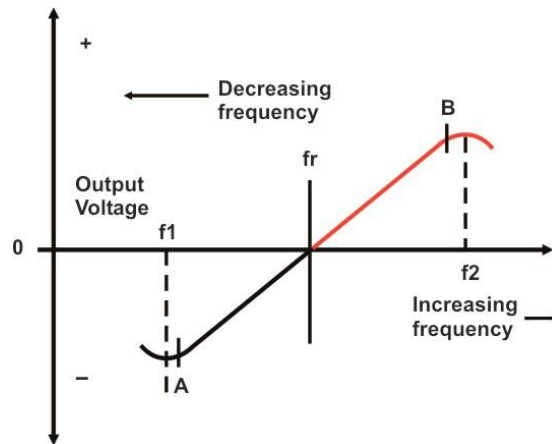


Fig. 10: Operation of Foster - Seeley Detector

### Operation of Phase-Locked Loop Detector

This is another demodulator that employs a phase comparator circuit. It is a very good demodulator and has the advantage that it is available, as a self-contained integrated circuit so there is no setting up required. You plug it in and it works. For these reasons, it is often used in commercial broadcast receivers. It has very low levels of distortion and is almost immune from external noise signals and provides very low levels of distortion.

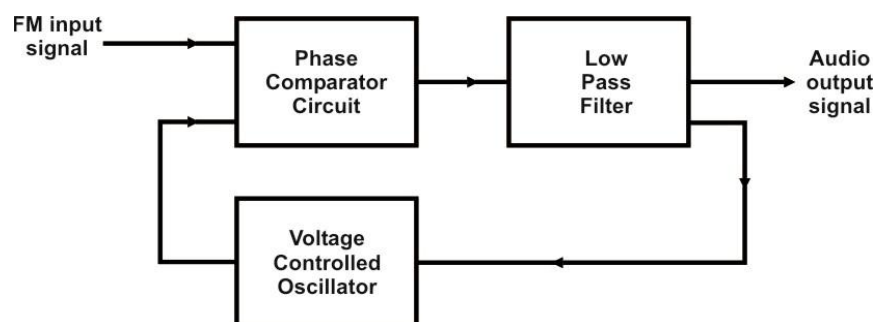


Fig. 11: Phase Lock Loop Detector

The DC output voltage from the output of the low pass filter, as shown in Fig. 11, controls the frequency of this oscillator. Now this DC voltage keeps the oscillator running at the same frequency as the original input signal and  $90^\circ$  out of phase. And if we did, then why not just add a phase shifting circuit at the input to give the  $90^\circ$  phase shift? The answer can be seen by imagining what happens when the input frequency changes - as it would with an FM signal. If the input frequency increases and decreases, the VCO

frequency is made to follow it. To do this, the input control voltage must increase and decrease. These changes in the DC voltage level form the demodulated signal. The signal then passes through a Low Pass Filter to smooth the output and then through an AC amplifier.

### Experimental Setup (Demodulation-1):

**Part B-1:** Frequency demodulation by Phase-Locked Loop Detector with Varactor modulator

#### Equipment:

1. IT-4103 Trainer Board
2. 2mm Patch Cords
3. CRO

#### Experimental Setup

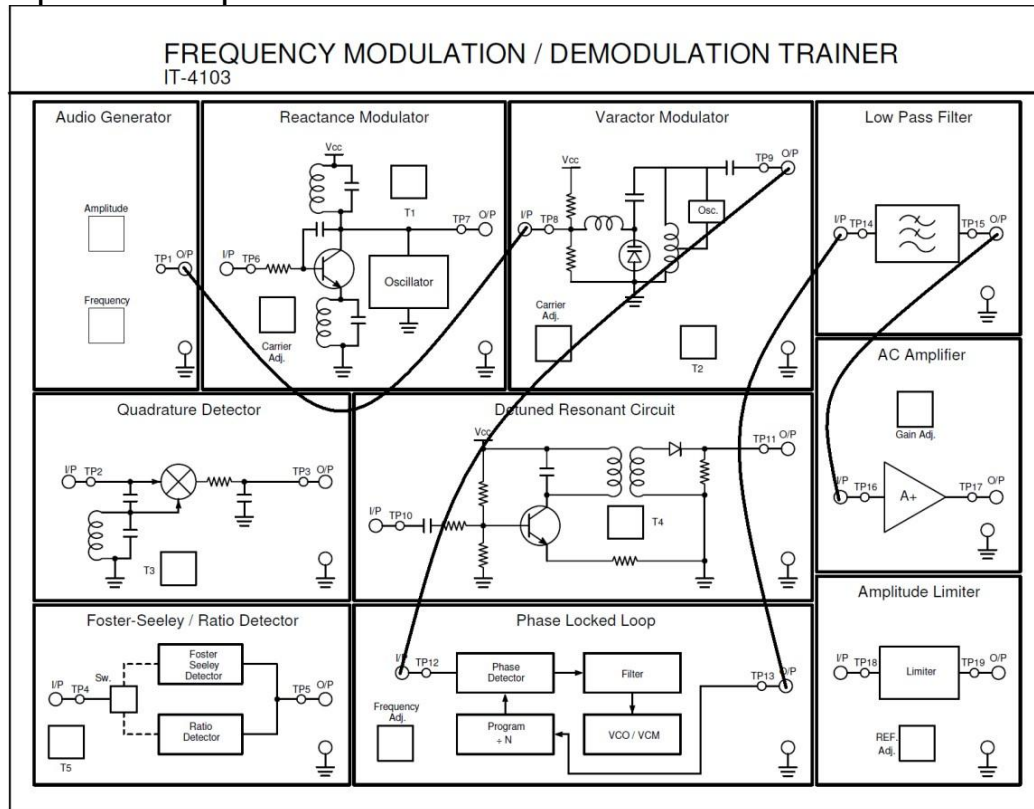


Fig. 12: Experimental Setup using Varactor Modulator and Phase-Locked Loop Detector

#### Procedure

Upon completion of this experiment students will be able to understand the concept of Phased Lock Loop Detector

1. Connect O/P of Audio Generator TP1 to Ch 1 of Oscilloscope. Check the O/P by adjusting Amplitude and Frequency knob to 2 KHz.
2. Connect O/P of Varactor Modulator TP8 to Ch 2 of Oscilloscope. Observe the signal at O/P and calculate the frequency of Carrier generated by this section. It should be around 420 KHz. If not so,

then using a plastic tuner, turn the screw of the Coil gently while observing the O/P at Oscilloscope.

3. Connect Audio O/P TP1 to Varactor Modulator Audio I/P TP8. Now connect O/P of Varactor Modulator TP9 to Ch 2 of Oscilloscope. Now the amplitude of the Audio signal varies. Variation in frequency as shown in Fig 13 can be observed.

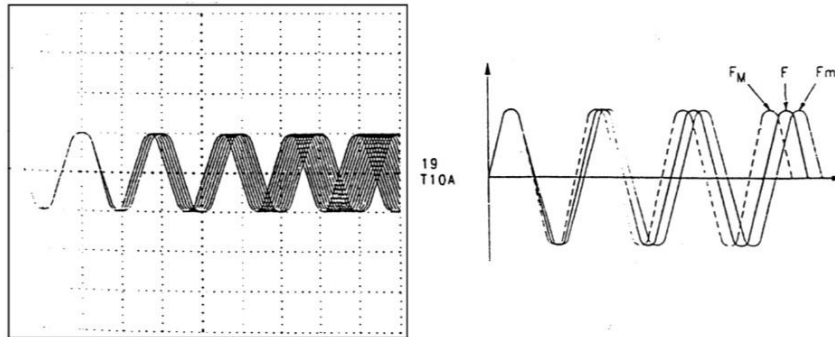
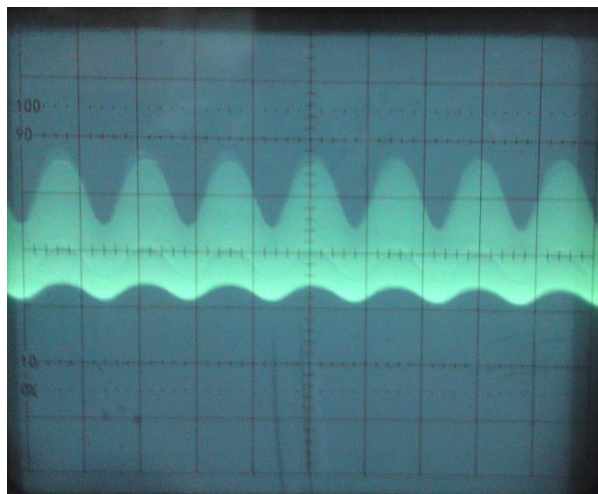


Fig 13: Frequency variation

4. Now connect the output of Varactor Modulator TP9 to the input of Phase Locked Loop TP12. Following is the output waveform of Phase Locked Loop.



5. If not observed; tune the IF coils of Varactor Modulator and Phase Locked Loop to acquire the above waveform.

**Note: The frequency tuned in procedure no. 2 is only the initialization frequency. The actual frequency is observed when we will obtain the audio.**

6. Connect the output of Phase Locked Loop TP13 to the Low Pass Filter and then to AC Amplifier.
7. Connect the O/P of AC Amplifier TP17 to Ch 2 of Oscilloscope. Note the effect of change in Amplitude of Audio signal on the Demodulated O/P signal.
8. If we increase the frequency the O/P signal will attenuate due limiting of Low Pass Filter.
9. Connect the Audio O/P to Ch 1 and compare this signal with the demodulated signal at O/P of AC Amplifier TP17.

### Experimental Setup (Demodulation-2):

#### Part B-2: Frequency demodulation by Foster - Seeley Detector with Reactance Modulator

##### Equipment:

4. IT-4103 Trainer Board
5. 2mm Patch Cords
6. CRO

##### Experimental Setup:

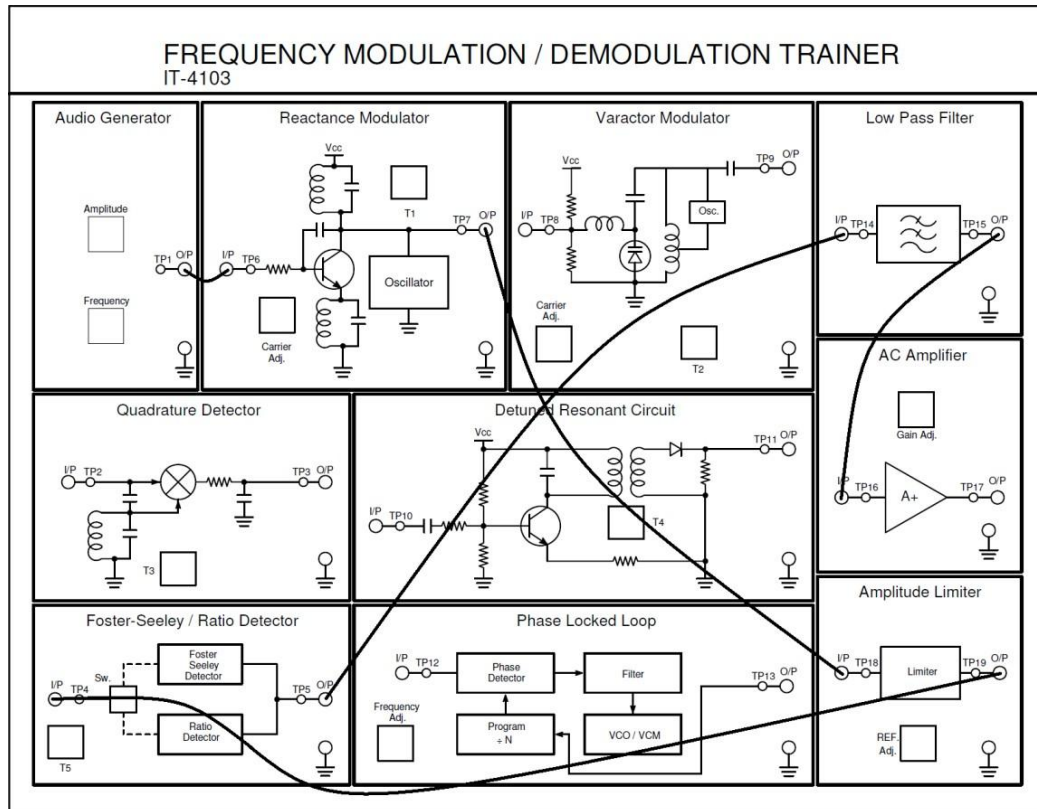
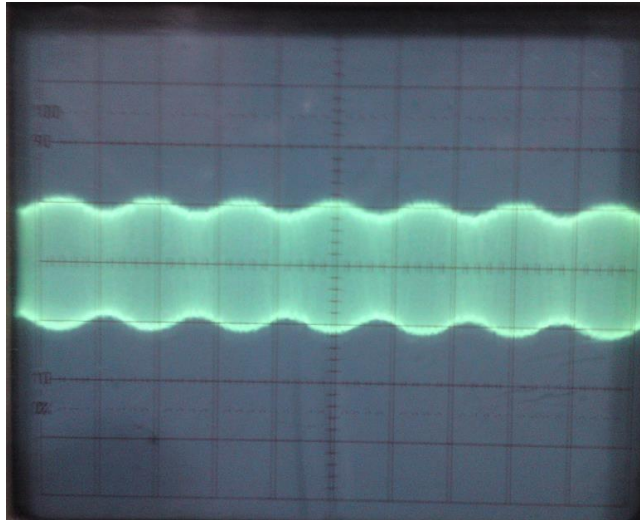


Fig. 14: Experimental Setup using Reactance modulator and Foster - Seeley Detector

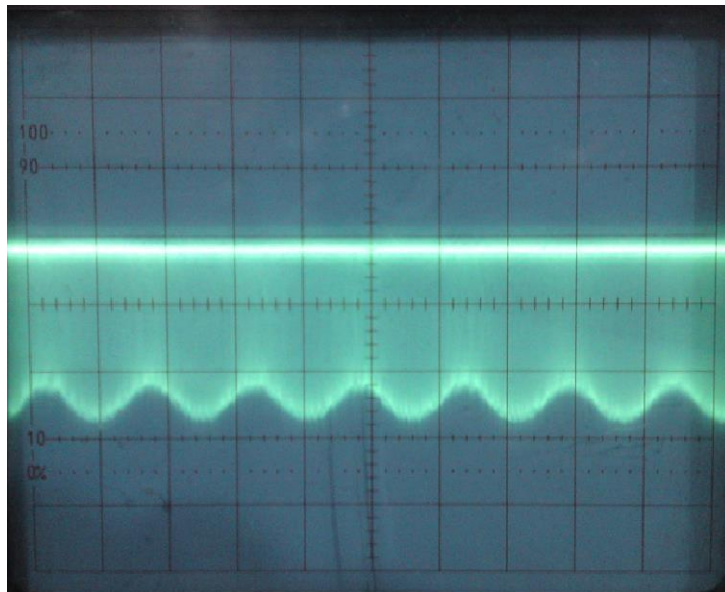
##### Procedure

Upon completion of this experiment students will be able to understand the concept of Foster Seeley Detector.

1. Connect O/P of Audio Generator TP1 to Ch 1 of Oscilloscope. Check the O/P by adjusting Amplitude and Frequency knob to 2 KHz.
2. Connect O/P of Reactance Modulator TP7 to Ch 2 of Oscilloscope. Observe the signal at O/P and calculate the frequency of Carrier generated by this section. It should around 360 KHz. If not so, then using a plastic tuner, turn the screw of the Coil gently while observing the O/P at Oscilloscope.
3. Connect Audio O/P TP1 to Reactance Modulator Audio I/P TP6. The following waveform will be observed.



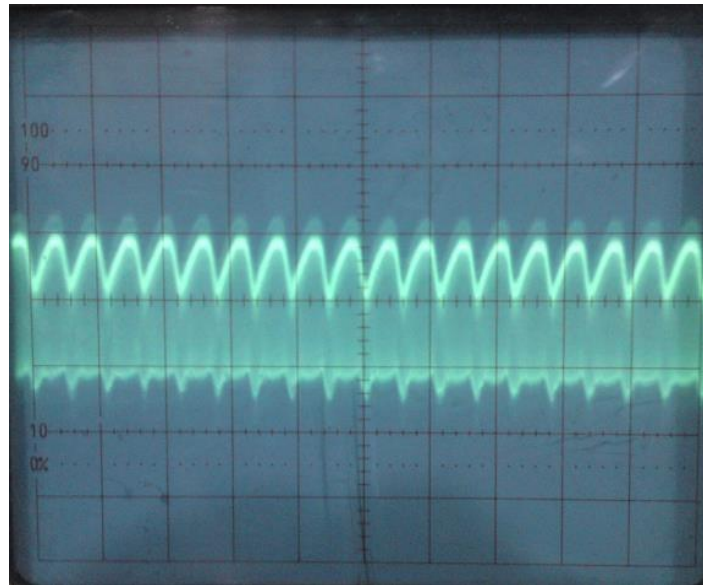
4. Now connect the O/P of Reactance Modulator TP7 to the input of Amplitude Limiter TP18. Connect output of Amplitude Limiter TP19 to Ch 2 of Oscilloscope. The following wave must be formed on your CRO after passing through Amplitude Limiter. Now the amplitude of the Audio signal varies. Variation in frequency can be observed.



**Note:** Vary the potentiometer of the Amplitude Limiter section so that the above waveform could be obtained.

5. Now connect the output of Amplitude Limiter TP19 to the input of Foster- Seeley/Ratio Detector TP4.
6. Make sure that Switch is in Foster Seeley Detector position. The following waveform may be observed.
7. If not observed; tune the IF coils of Reactance Modulator and Foster- Seeley/Ratio Detector to acquire the above waveform.

**Note:** The frequency tuned in procedure no. 2 is only the initialization frequency. The actual frequency is observed when we will obtain the audio.



8. Connect the output of Foster-Seeley/Ratio Detector TP5 to the Low Pass Filter and then to AC Amplifier.
9. Connect the O/P of AC Amplifier to Ch 2 of Oscilloscope. Note the effect of change in Amplitude of Audio signal on the Demodulated O/P signal.
10. If we increase the frequency the O/P signal will attenuate due to limiting of Low Pass Filter.
11. Connect the Audio O/P to Ch 1 and compare this signal with the demodulated signal at O/P of AC Amplifier.

**Report:**

- 1) Describe the principle of operation of the VARACTOR MODULATOR and the FOSTER-SEELEY DETECTOR.
- 2) Show all the observations with necessary wave shapes in your report and describe them.



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Experiment No: 5 (Hardware)

Experiment Name: - Sampling and Signal Reconstruction

**Objectives:**

1. Investigate methods used to sample a signal and to recognize the signal that results from each method.
2. Investigate a method used to reconstruct the intelligence from a sampled signal and demonstrate how the sampling signal frequency and the low pass filter characteristic affect reconstruction.

**Equipment and Materials:**

- 1) IT-4201 module
- 2) 2mm Patch Cords
- 3) CRO

**Theoretical Background:**

**Theory of Sampling**

In analog communication systems like AM, FM, the instantaneous value of the information signal is used to hang certain parameter of the carrier signal.

Pulse modulation systems differ from these systems in a way that they transmit a limited number of discrete states of a signal at a predetermined time; sampling can be defined as measuring the value of an information signal at predetermined time intervals. The rate at which the signal is sampled is known as the sampling rate or sampling frequency.

It is the major parameter, which decides the quality of the reproduced signal. If the signal is sampled quite frequently (whose limit is specified by Nyquist Criterion), then it can be reproduced exactly at the receiver with no distortion.

**Pulse Amplitude Modulation**

The pulse amplitude modulation (PAM) system is analog system where the train of Pulses corresponding to the samples of each signal are modulated in amplitude in accordance with the signal itself i.e. the height of the transmitted pulses vary with the amplitude of the message. (Fig.1 & 2)

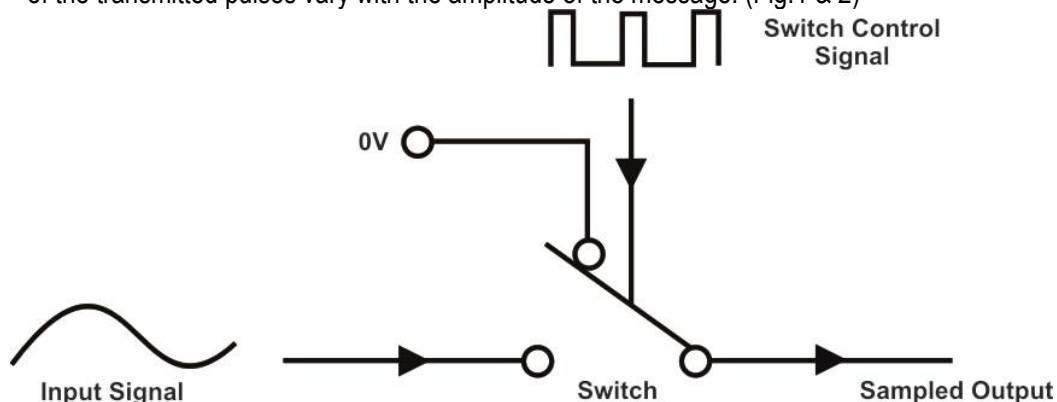


Fig.1: Basic Sampling Process

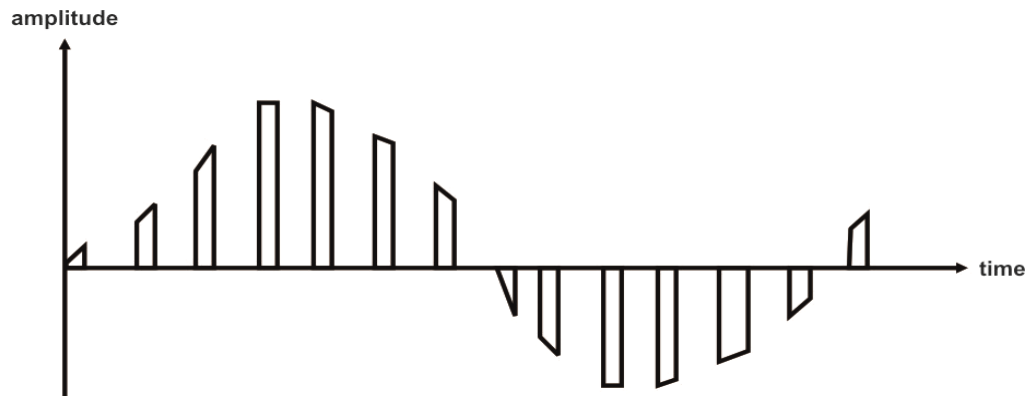


Fig. 2: Sampled Output

PAM is an analog system because the amplitude of pulse can vary infinitely i.e. the levels is not discrete. An information signal sent through an ideal switch, which is operated, by a control signal, isolated from the information signal, produces a PAM signal. When the switch is open, the voltage is zero; when switch is closed, the output voltage is equal to the instantaneous signal voltage. The sample width depends upon how long a switch remains closed.

**Note:** An ideal switch is free of delay and contact bounce problems.

### Nyquist Criterion (Sampling Theorem)

The Nyquist criterion states that a continuous signal band limited to  $F_m$  Hz can be completely represented by and reconstructed from the samples taken at a rate greater than or equal to  $2F_m$  samples/second. This minimum sampling frequency is called as Nyquist Rate i.e. for faithful reproduction of information signal  $f_s > 2 f_m$ .

### Effect of Sample and Sample / Hold Outputs

If the pulse width of the carrier pulse train used in natural sampling is made very short compared to the pulse period, the natural PAM is referred to as instantaneous PAM. As it has been discussed, shorter pulse is desirable for allowing many signals to be included in TDM format but the pulse can be highly corrupted by noise due to lesser signal power.

One way to maintain reasonable pulse energy is to hold the sample value until the next sample is taken. This technique is termed as sample-and-hold techniques. The sample-and-hold waveform looks as shown in Fig. 3.

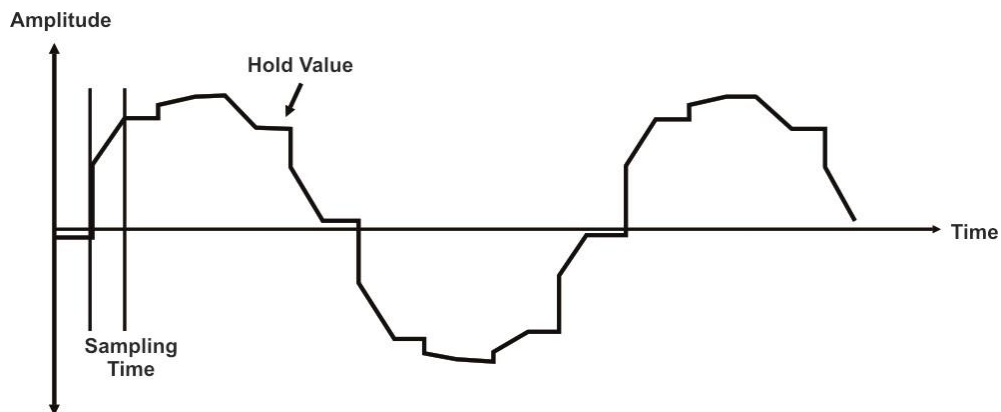


Fig. 3: Sample & Hold Waveform

Now, the area under the curve (which is equivalent to the signal power) is greater and so the filter output amplitude and quality of reproduced signal is improved. The 'hold' facility can be provided by a capacitor when the switch connects the capacitor to PAM output it charges to the instantaneous value.

### Aliasing

If the signal is sampled at a rate lower than stated by Nyquist criterion, then there is an overlap between the information signal and the sidebands of the harmonics. Thus, the higher and the lower frequency components get mixed and cause unwanted signals to appear at the demodulator output. This phenomenon is termed as aliasing or fold over distortion. The various reasons for aliasing and its prevention are as described.

1. Aliasing due to Under-Sampling.
2. Aliasing Due to Wide Band Signal.
3. Aliasing Due to Filter Roll-off.
4. Aliasing due to Noise.

### Low Pass Filters

The PAM system the message is recovered by a low pass filter. The type of filter used is very important, as the signal above the cut-off frequency would affect the recovered signal if they are not attenuated sufficiently.

#### Filter Basic:

The simplest type of filter is a resistance-capacitance (RC) filter. The pass filter and low pass RC filters are as shown in Fig. 4. The analysis of these filters becomes easier if we think of them as a.c. potential dividers. The reactance of the capacitor is frequency dependent with a high value at low frequencies and a low value at high frequencies.

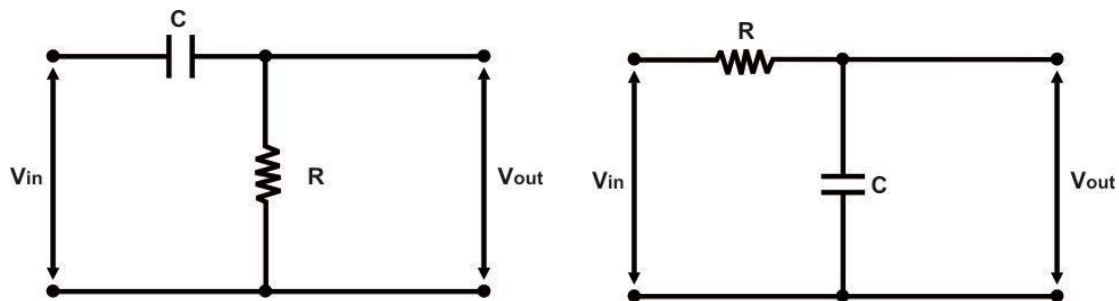


Fig. 4: (a): Passive High Pass Filter and (b): Passive Low Pass Filter

In case of high pass filter, the series capacitance has high reactance at low frequencies and hence results in reduction in output voltage. An increase in frequency causes an increase in output voltage with  $V_{OUT}$  approaching input voltage  $V_{IN}$ . The effect of capacitance is just opposite case of low pass filter. Here, the capacitance is in short and hence  $V_{OUT}$  reduces as frequency increases thereby decreasing its reactance.

## Part A: Signal Sampling and Reconstruction techniques (Sample Only)

### Experimental Setup:

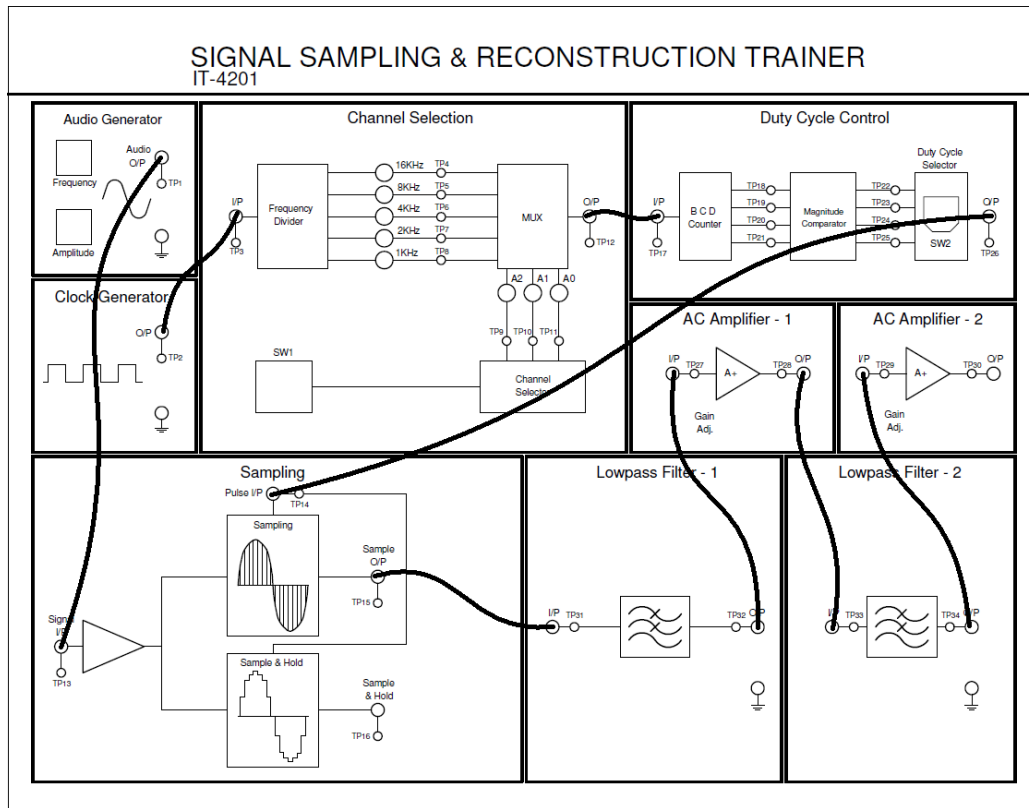


Fig. A.1: Experimental Setup

### Procedure:

1. Turn on the power of IT-4201 module.
2. Adjust Frequency Adj of Audio Generator to approximately 200 Hz.
3. Adjust Amplitude Adj Knob of Audio Generator in center position.
4. Connect Oscilloscope probe to Audio O/P TP1.
5. Make connections using 2mm patch cords as shown in Fig. A.1.
6. Select the 16 kHz channel.
7. Select minimum Duty Cycle.
8. Connect CH1 to Sample O/P TP15. The display shows 200 Hz sine wave being sampled at 1.6 KHz.

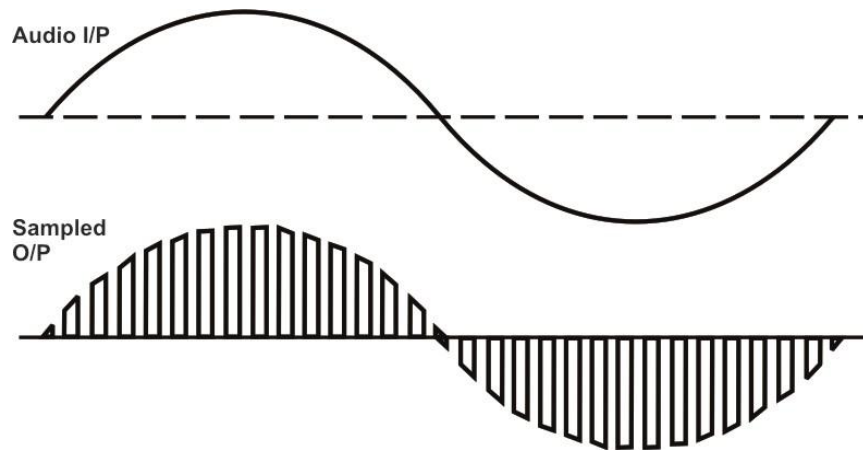


Fig. A.2: Sampled Signal

9. By successive presses of Frequency Selector switch, change the sampling frequency to 2KHz, 4KHz, 8KHz, and back to 16KHz. Observe how SAMPLE output changes in each case.
10. Connect Sampled O/P TP15 to LPF 1 I/P TP31 and connect CH2 to LPF 1 O/P TP31. Change the sampling frequency by pressing Frequency Selector switch and observe how at lower sampling frequencies introduce distortion into the filter output waveform.
11. Connect the output of LPF 1 TP32 to the input of amplifier 1 TP27. The amplified output may be observed.
12. Now connect Amplifier 1 output TP28 to the input of LPF 2 TP33. Connect CH1 to Signal I/P TP13 and connect CH2 to LPF 2 O/P TP34 and observe a clear Audio signal.
13. Connect the output of LPF 2 TP34 to the input of Amplifier 2 TP29 to observe the amplified output.

## Part B: Signal Sampling and Reconstruction techniques (Sample & Hold Signal)

### Experimental Setup:

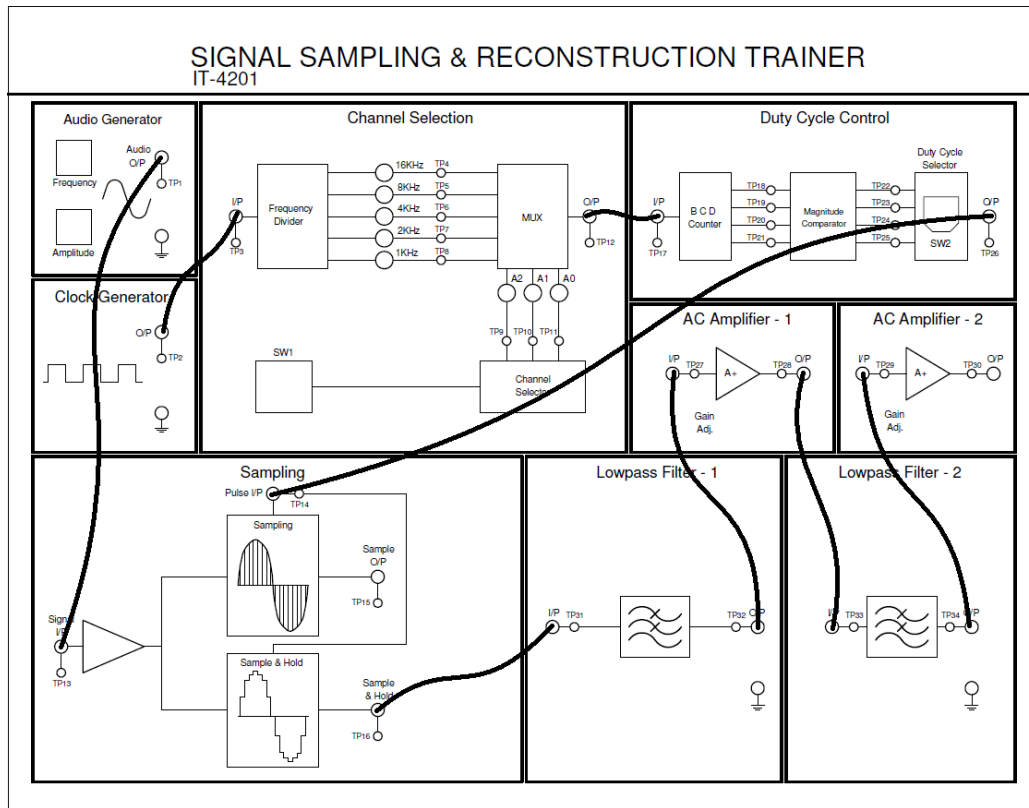


Fig. B.1: Experimental Setup

### Procedure:

1. Turn on the power of IT-4201 module.
2. Adjust Frequency Adj of Audio Generator to approximately 200 Hz.
3. Adjust Amplitude Adj Knob of Audio Generator in center position.
4. Connect Oscilloscope probe to Audio O/P TP1.
5. Make connections using 2mm patch cords as shown in Fig. B.1.
6. Select the 16 kHz channel.
7. Select minimum Duty Cycle.
8. Connect CH1 to Sample & Hold O/P TP16. The display shows Sine wave being Sample & Hold at 1.6KHz.
9. By successive presses of Frequency Selector switch, change the sampling frequency to 2KHz, 4KHz, 8KHz, and back to 16KHz. Observe how the O/P changes with respect to change in Sampling Frequency.
10. Vary the position of Duty Cycle Selector switch from 0% to 90% (position 0 to 9), Observing how the Sample & Hold O/P changes with respect to Duty Cycle change.
11. Connect Sampled & Hold O/P TP16 to LPF 1 I/P TP31 and connect CH2 to LPF 1 O/P TP32.

Change the sampling frequency by pressing Frequency Selector switch, and observe how the lower sampling frequencies introduce distortion into the filter output waveform.

12. Connect the output of LPF 1 TP32 to the input of amplifier 1 TP27. The amplified output may be observed.
13. Now connect Amplifier 1 output TP28 to the input of LPF 2 TP33. Connect CH1 to Signal I/P TP13 and connect CH2 to LPF 2 O/P TP34 and observe a clear Audio signal.
14. Connect the output of LPF 2 TP34 to the input of Amplifier 2 TP29 to observe the amplified output.
15. Remove all the patch cords from the trainer and switch off power supply.

### Part C: Signal Sampling and Reconstruction techniques (Sampling and Sample & Hold Signal)

#### Experimental Setup:

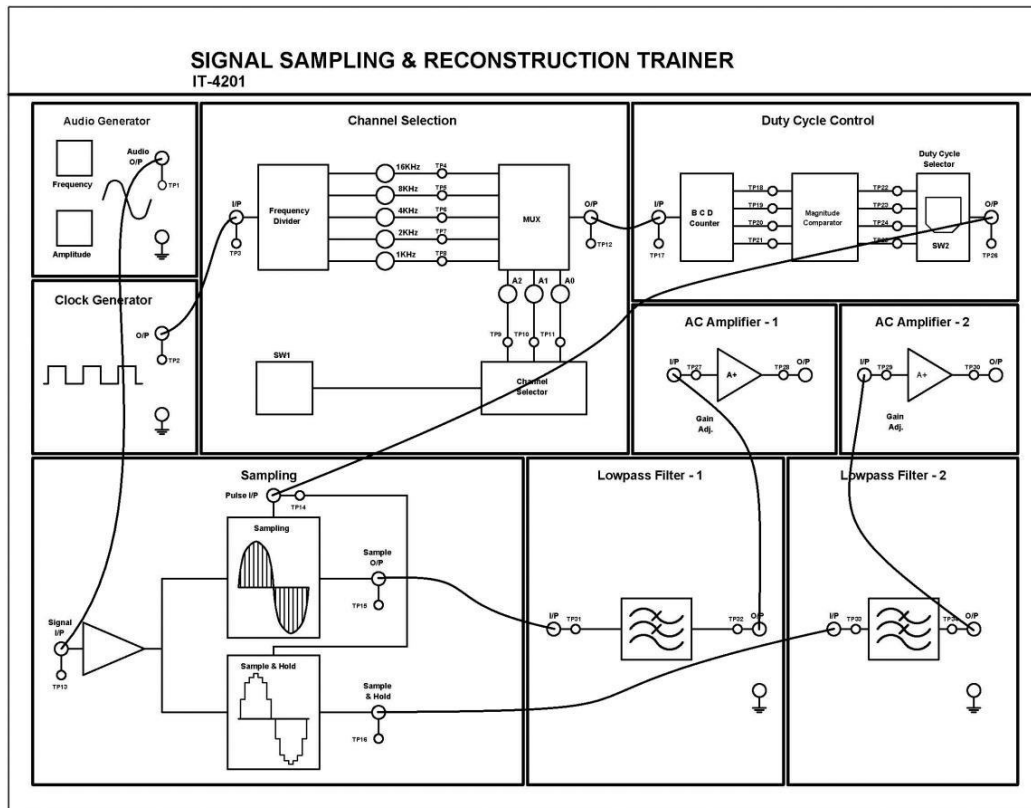


Fig. C.1: Experimental Setup

#### Procedure:

1. Turn on the power of IT-4201 module.
2. Adjust Frequency Adj of Audio Generator to approximately 200 Hz.
3. Adjust Amplitude Adj Knob of Audio Generator in center position.
4. Connect Oscilloscope probe to Audio O/P TP1.
5. Make connections using 2mm patch cords as shown in Fig. C.1.

6. Select the 16 kHz channel.
7. Select minimum Duty Cycle.
8. Connect CH1 to Sample O/P TP15. The display shows 200 Hz sine wave being sampled at 1.6 KHz.
9. Now Connect CH2 to Sample & Hold O/P TP16. The display shows Sine wave being Sample & Hold at 1.6KHz.
10. Observe the difference.
11. By successive presses of Frequency Selector switch, change the sampling frequency to 2KHz, 4KHz, 8KHz, and back to 16KHz. Observe how SAMPLE output changes in each case.
12. Connect Sampled O/P TP15 to LPF 1 I/P TP31 and LPF-1 TP32 to AC Amplifier- 1 I/P TP27. Connect CH1 to TP28.
13. Connect Sample & Hold O/P TP17 to LPF-2 I/P TP33 and LPF-2 O/P TP34 to AC Amplifier-2 I/P TP29. Connect CH2 to TP30.
14. Observe the Outputs.
15. Remove all the patch cord from the trainer and switch off power supply.



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Experiment No: 4 (Software)

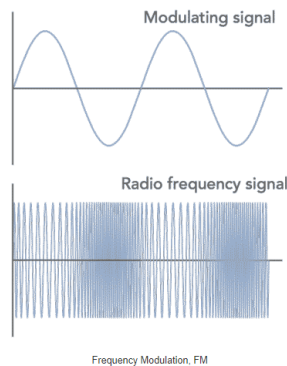
Experiment Name:- **Frequency Modulation & Demodulation using MATLAB Simulink**

**Objective:** To perform the Frequency Modulation signal Generation and Detection using Matlab Simulink.

**Theoretical Background:**

**Frequency Modulation-** To generate a frequency modulated signal, the frequency of the radio carrier is changed in line with the amplitude of the incoming audio signal.

When the audio signal is modulated onto the radio frequency carrier, the new radio frequency signal moves up and down in frequency. The amount by which the signal moves up and down is important. It is known as the deviation and is normally quoted as the number of kilohertz deviation. As an example the signal may have a deviation of plus and minus 3 kHz, i.e.  $\pm 3$  kHz. In this case the carrier is made to move up and down by 3 kHz.



Broadcast stations in the VHF portion of the frequency spectrum between 88.5 and 108 MHz use large values of deviation, typically  $\pm 75$  kHz. This is known as wide-band FM (WBFM). These signals are capable of supporting high quality transmissions, but occupy a large amount of bandwidth. Usually 200 kHz is allowed for each wide-band FM transmission. For communications purposes less bandwidth is used. Narrow band FM (NBFM) often uses deviation figures of around  $\pm 3$  kHz.

It is narrow band FM that is typically used for two-way radio communication applications. Having a narrower band it is not able to provide the high quality of the wideband transmissions, but this is not needed for applications such as mobile radio communication.

**Frequency demodulation-** As with any form of modulation, it is necessary to be able to successfully demodulate it and recover the original signal. The FM demodulator may be called a variety of names including FM demodulator, FM detector or an FM discriminator. There are a number of different types of FM demodulator, but all of them enable the frequency variations of the incoming signal to be converted into

amplitude variations on the output. These are typically fed into an audio amplifier, or possibly a digital interface if data is being passed over the system.

### **Software and Device requirement:**

MATLAB (with Simulink)

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit); Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

### **Problem Statement:**

- Draw all the models one-by-one in the MATLAB Simulink window.
- Set the simulation time from the model window as below:  
*Simulation▢ Simulation parameters: Start time = 0.0 & Stop time = 50.0.*
- Then simulate all the models and observe all the wave shapes at different points of the model.

### **Parameters to be used for different blocks-**

☐ **Sine wave:**

- ☐ *Sine type: Time based*
- Amplitude: 1.0*
- Frequency: 2rad/sec*
- Sample time: 0.001*

☐ **Constant:** *Constant value: 20*

☐ **Gain:** *Gain value: 10 & Multiplication: Element wise*

☐ **Gain1:** *Gain value: 2\*pi & Multiplication: Element wise*

☐ **Integrator:** *external reset: none, initial condition source: internal, **select** enable zero-crossing detection.*

☐ **Product:** *Number of inputs: 2 & Multiplication: Element wise*

☐ **Product:** *Number of inputs: 2*

☐ **Trigonometric Function:** *function: cos, approximation method: none, output signal type: auto.*

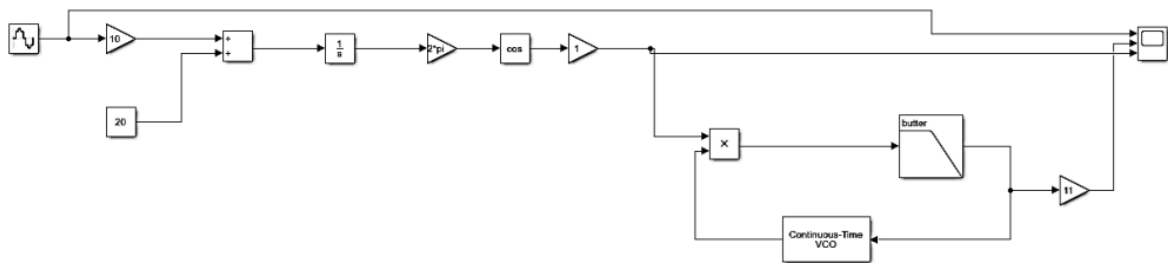
☐ **Analog Filter design:** *design method: Butterworth, filter type: low pass, filter order: 10, passband frequency: 2*

- **Continuous Time VCO:** output amplitude: 1, quiescent frequency: 20, input sensitivity:  $10/(2\pi)$ , phase =  $\pi/2$
- **Sum:** List of signs: |+- (subtraction will be done). Sample time: 0.001

Note: To sum up two inputs, list of signs: |++

## Simulation Model

$$s(t) = A_c \cos \left[ 2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau \right]$$



## Lab Report:

- i. Design FM Modulation and Demodulation circuits using Built-in blocks of MATLAB
- ii. Write down a short paragraph explaining the functionalities of VCO and how Phase Locked Loop (PLL) method of demodulation works.



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**Experiment No: 5 (Software)**

**Experiment Name:- Sampling Theorem using MATLAB Simulink**

**Objective:** The objective of this experiment is to study the Sampling Theorem.

**Theoretical Background:**

Given a continuous sinusoidal signal  $s(t)$  and an impulse train  $p(t)$ , we can sample continuous signal  $s(t)$  by multiplying with  $p(t)$ . Suppose the period of impulses in  $p(t)$  is  $T_s$  and its corresponding sampling frequency will be  $1/T_s = F_s$ . In addition, the continuous sinusoidal signal has frequency  $F$ .

In this section, we will discuss three cases,

- $F_s < 2F$  (Sampling rate is less than twice sine wave frequency)
- $F_s = 2F$  (Sampling rate is equal to twice sine wave frequency)
- $F_s > 2F$  (Sampling rate is larger than twice sine wave frequency)

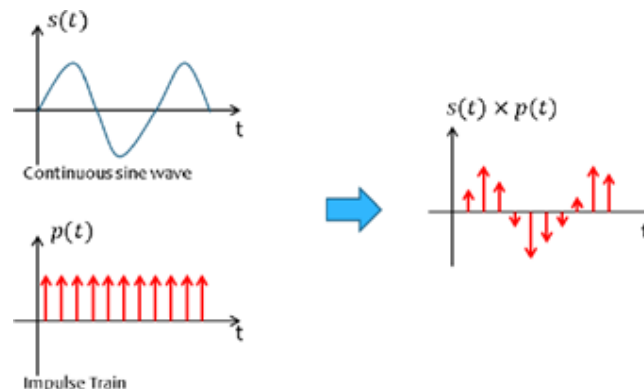


Fig. 1: Sampling Theorem

From sampling theorem, sampling rate  $F_s$  should be equal or larger than twice frequency of sinusoidal signal  $F$ . i.e.,  $F_s \geq 2F$ . For our Simulink configuration, we set the sinusoidal signal frequency as 500Hz and we give three sampling frequencies, 500Hz, 1kHz and 10kHz. For Nyquist theorem, if the sampling rate is smaller, it will result in aliasing. If the sampling rate is larger than the Nyquist rate, it will result in imaging. Hence, in this section, we want to give a true simulation of sampling a continuous sinusoidal signal and give discussion in "Time Domain".

**Software and Device requirement:**

MATLAB (with simulink)

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit);

Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

**Problem Statement:**

1. Draw all the models one-by-one in the *MATLAB Simulink* window.
2. Set the **simulation time** from the model window as below:

*Simulation → Simulation parameters: Start time = 0.0 & Stop time = 0.01.*

3. Simulate the model and observe all the wave shapes at different points of the model.

**Procedure:**

1. You can take the blocks from the *Simulink Library Browser*.
  - a. The blocks that you will use in this experiment can be taken from the following mentioned library:
    - I. Sine Wave: *Simulink → Sources → Sine wave*
    - II. Pulse Generator: *Simulink → Sources → Pulse Generator*
    - III. Product: *Simulink → Math operations → Product*
    - IV. Scope: *Simulink → Sinks → Scope*

**Parameters to be used for different blocks:**

**1. Sine wave:**

*Sine type: Time based  
Amplitude: 5.0  
Frequency:  $2\pi \times 500$  rad/sec  
Sample time: 1/1000000*

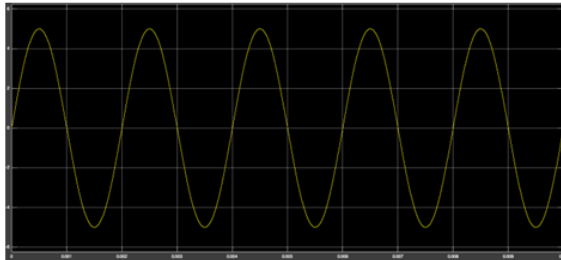
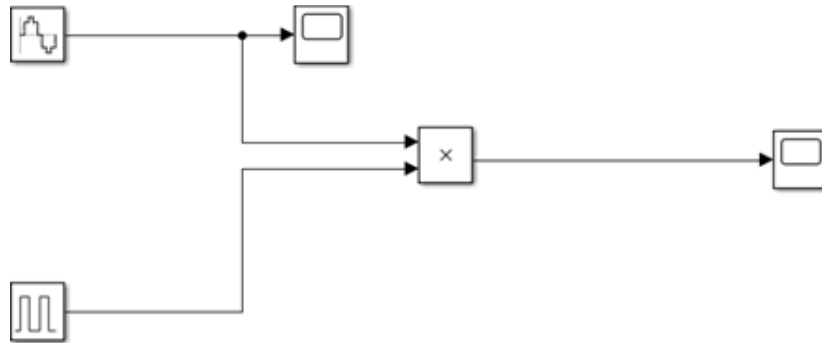
**2. Pulse Generator:**

*Pulse type: Time based  
Amplitude: 1.0  
Period (secs): 0.0001  
Pulse width (% of period): 0.1*

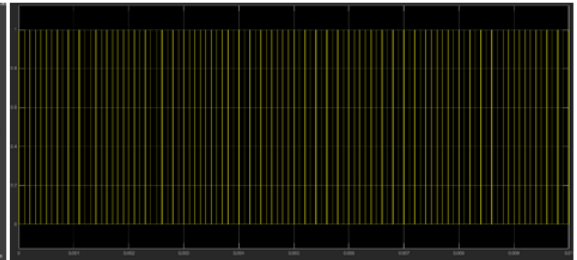
**3. Product: Number of inputs: 2 & Multiplication: Element wise**

**4. Scope: Data History: Uncheck the 'Limit data points to last'.**

## Simulation Models

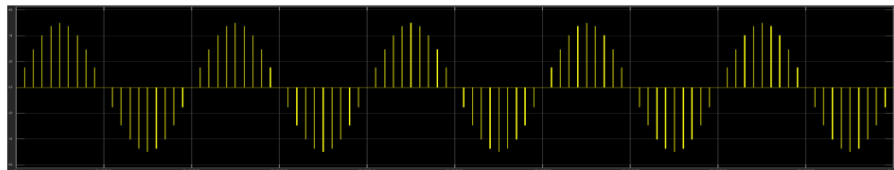


*Fig: Sine wave*



*Fig: Impulse Train*

Output-



*Fig: Sampled Signal*

## Report:

1. At first regenerate the model shown in the experiment. Then build the model for reconstruction on top of that. Show all the wave shapes in the report of the full model.
2. Explore natural sampling and show the wave shapes in the report.



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Experiment No: 6 (Hardware)

Experiment Name: - Amplitude Shift-Keying

**Theoretical Background:**

**Amplitude-shift keying (ASK)** is a form of modulation that represents digital data as variations in the amplitude of a carrier wave.

The amplitude of an analog carrier signal varies in accordance with the bit stream (modulating signal), keeping frequency and phase constant. The level of amplitude can be used to represent binary logic 0s and 1s. We can think of a carrier signal as an ON or OFF switch. In the modulated signal, logic 0 is represented by the absence of a carrier, thus giving OFF/ON keying operation and hence the name given.

Like AM, ASK is also linear and sensitive to atmospheric noise, distortions, propagation conditions on different routes. Both ASK modulation and demodulation processes are relatively inexpensive.

The simplest and most common form of ASK operates as a switch, using the presence of a carrier wave to indicate a binary one and its absence to indicate a binary zero. This type of modulation is called **on-off keying** as shown in Fig. 1.

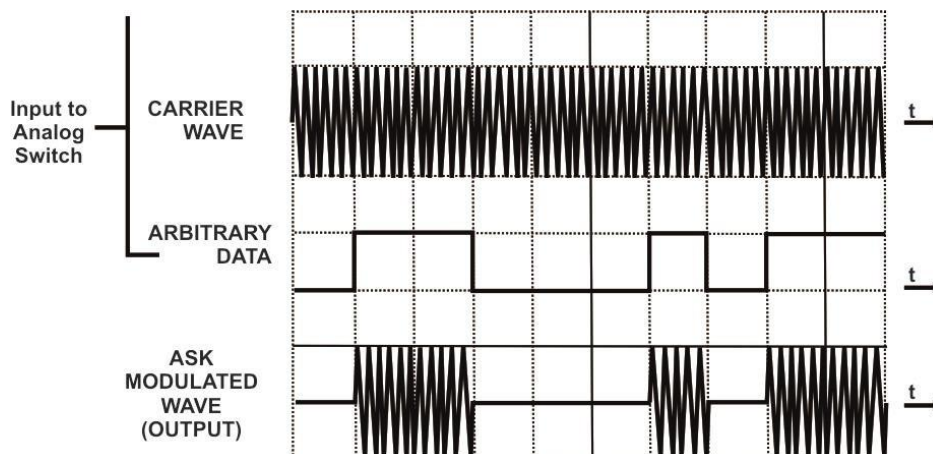


Fig. 1: ASK modulation

The method to demodulate the ASK modulation results in a great simplicity at the receiver. The method to demodulate the ASK waveform is to a system which, by using VCO generates the Carrier frequency of ASK. When it receives an in band Carrier signal from ASK input, it saturates thus causing a logic '0'. Similarly when it does not receive any Carrier signal from ASK input it goes into cutoff causing a logic '1'. In this way we will receive an inverted Data at DATA O/P terminal as shown in Fig. 2.

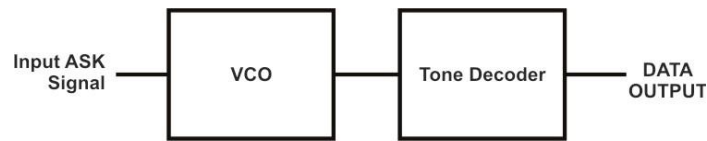


Fig. 2: ASK Demodulator

### Part A: ASK Modulation

#### Instrument:

1. IT-4206 Trainer Board
2. 2mm Patch Cords
3. CRO

#### Experimental Setup:

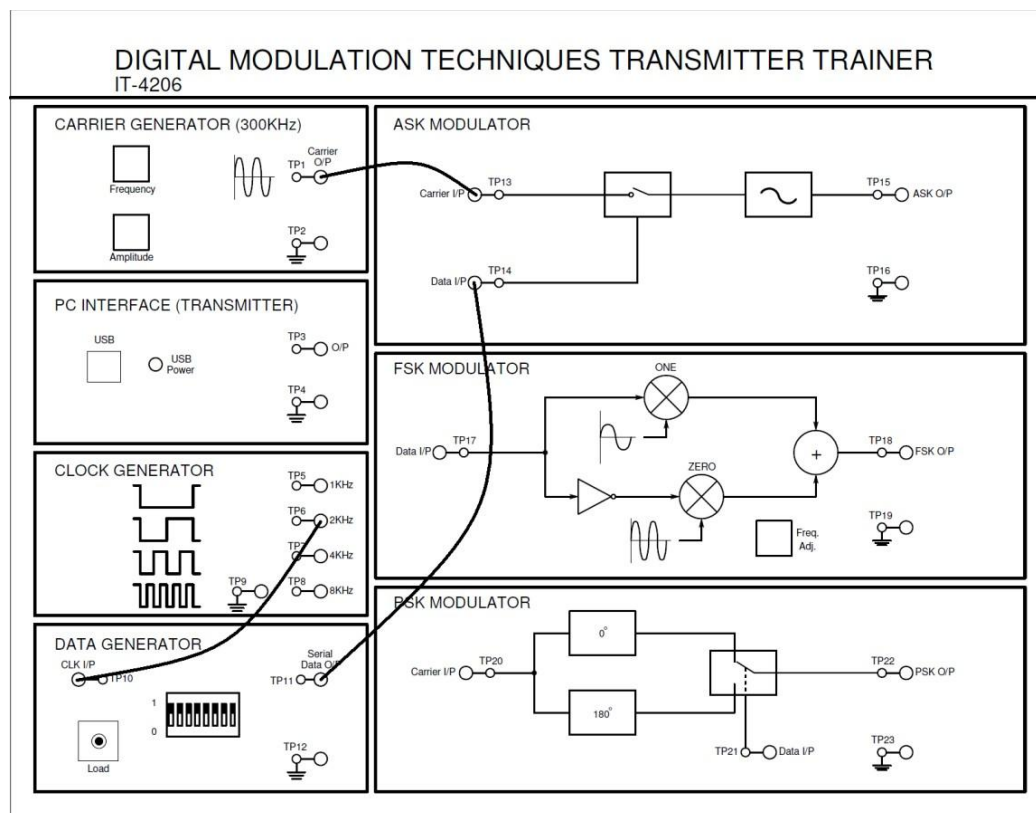


Fig. A.1: Experimental Setup

#### Procedure

1. Connect Oscilloscope probe to Carrier O/P test point TP1 and set the Frequency, using Frequency Adj. knob to 300 KHz.
2. Provide 2 KHz clock TP6 to the clock input of Data Generator TP11.
3. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved. 1 1 0 0 1 1 0
4. After setting the DIP switches press 'Load' switch.
5. Connect the Carrier O/P TP1 to TP13 and Serial Data O/P TP11 to TP14 in ASK Modulator

Block.

6. Connect CH1 to Data O/P test point TP11 and CH2 to ASK O/P test point TP15.
7. Verify the ASK Modulation waveform with respect to Data O/P.
8. Different Data Patterns can be obtained and experimented by changing DIP switches and pressing Load switch.

## Part B: ASK Demodulation

### Instrument:

1. IT-4206 Trainer Board
2. IT-4207 Trainer Board
3. 2mm Patch Cords
4. CRO

### Experimental Setup:

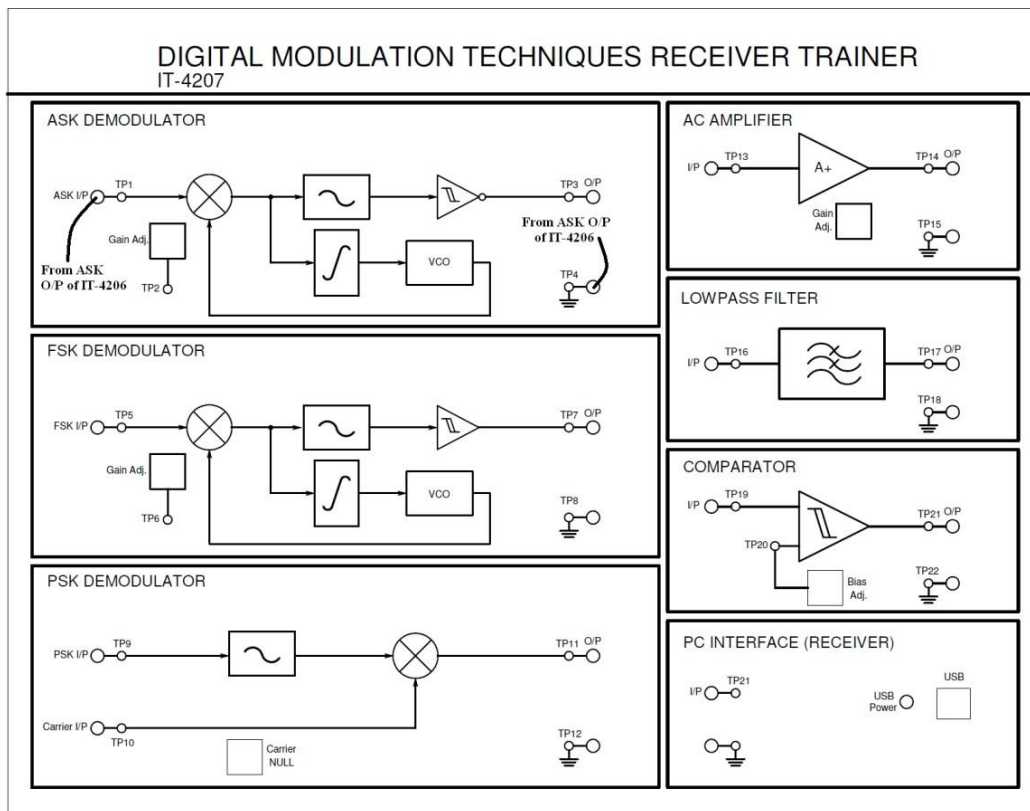


Fig B.1: Experimental Setup

### Procedure

1. Make connections using 2mm patch cords as shown in Fig B.1 using IT-4206.
2. Connect Oscilloscope probe to Carrier O/P test point TP1 of Module IT-4206 and set the Frequency, using Frequency Adj. knob to 300 KHz.
3. Provide 2 KHz clock TP6 to the clock input of Data Generator TP10.

4. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved. 1 1 1 1 1 1 1 1
5. After setting the DIP switches press 'Load' switch.
6. Carrier can be observed at ASK O/P test point TP15 using CH 1 of Oscilloscope.
7. Turn on the power of IT-4207 module.
8. Connect GND terminal of IT-4206 and IT-4207 using 2mm patch cord.
9. Set the CH2 of Oscilloscope to DC Coupling.
10. Connect ASK O/P TP15 of IT-4206 to ASK I/P TP1 of IT-4207, using 2mm patch cord, and observe the Data O/P test point TP3 of IT-4207 using CH2 of Oscilloscope as shown in Fig 3.1
11. Data O/P TP3 of IT-4207 should show DC Zero Level, if not then bring the DATA O/P to 'Zero Level' by adjusting the Carrier Frequency, using Frequency Adj. knob of IT-4206 and by adjusting the Gain Knob of IT-4207.
12. By achieving the 'Zero Level' at Data O/P TP3 of IT-4207 we have correctly tuned the Tone Decoder to the Carrier Frequency of the ASK Modulator. Now we can receive the ASK Modulated Data correctly.
13. Now adjust the DIP Switches of IT-4206, starting from left, so that following data sequence is achieved. 1 1 0 0 0 1 1 0
14. After setting the DIP switches press 'Load' switch
15. Observe the change in ASK O/P TP15 of IT-4206 using CH 1 of Oscilloscope
16. Now connect CH1 of Oscilloscope to Serial Data O/P TP11 of IT-4206 and Data O/P test point TP3 of IT-4207 to CH2. Verify that the Data Modulated, using ASK Modulation, and Data Received, using ASK De-Modulation are same.



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Experiment No: 7 (Hardware)

Experiment Name: - Frequency Shift-Keying

**Theoretical Background:**

In **frequency shift keying**, the carrier frequency is shifted in steps corresponding to the digital modulation signal. If we use high frequency to represent a Data '1' & low frequency to represent Data '0', the resulting Frequency shift keying waveform appears as shown in Fig. 1.

Thus    Data = 1 high frequency  
          Data = 0 low frequency

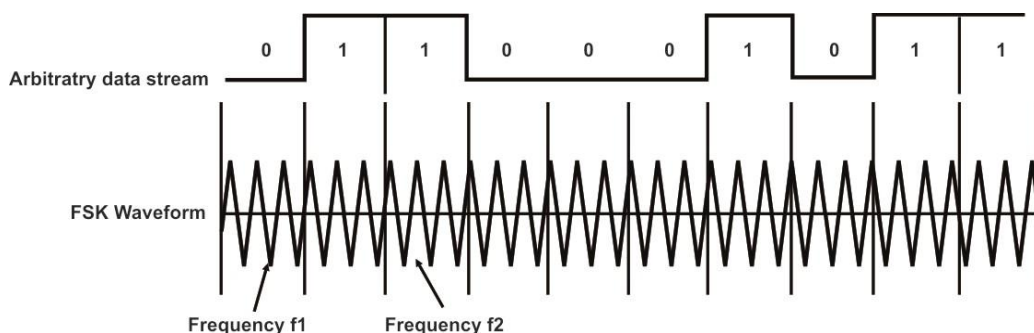


Fig. 1: FSK Waveform

FSK waveform, can be represented as the sum of two ASK waveforms, as shown in Fig. 2.

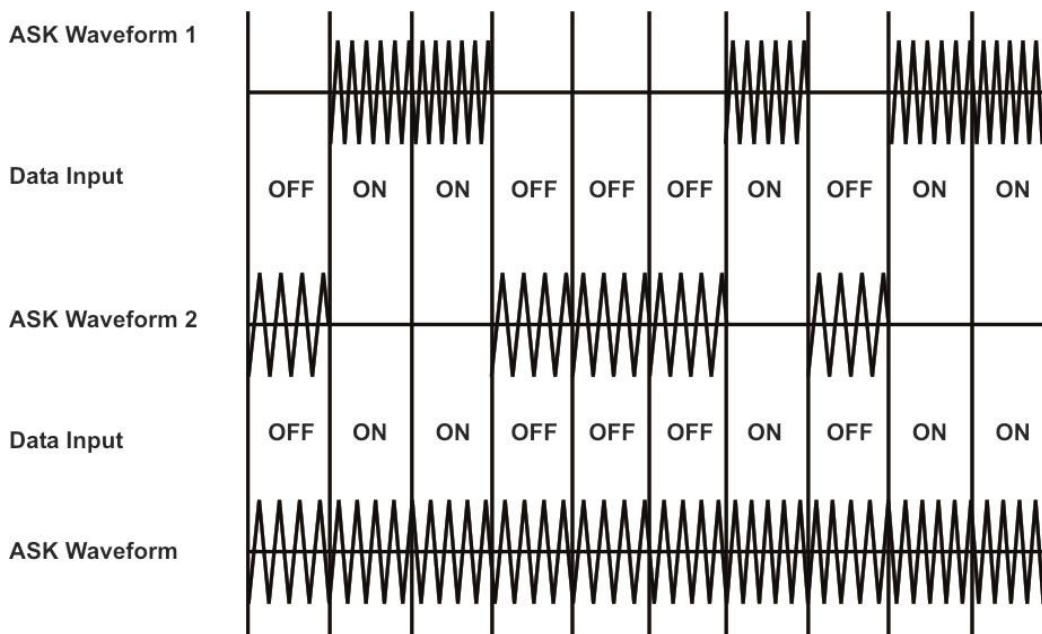


Fig. 2: Generation of FSK Waveform from ASK Waveforms

We choose an arbitrary Data Stream as given below. In our case, as we have observed in ASK Modulator that we receive inverted Data after ASK De-Modulation. So we will use Low Frequency for Data '1' and High Frequency for Data '0'.

*Data Stream*

Original Data Stream: 1 0 0 1 1 1 0 1

Inverted Data Stream: 0 1 1 0 0 0 1 0

We can apply the ASK technique devised to our Data Stream as shown in Fig. 3 Data '0' = High Frequency

Data '1' = Low Frequency

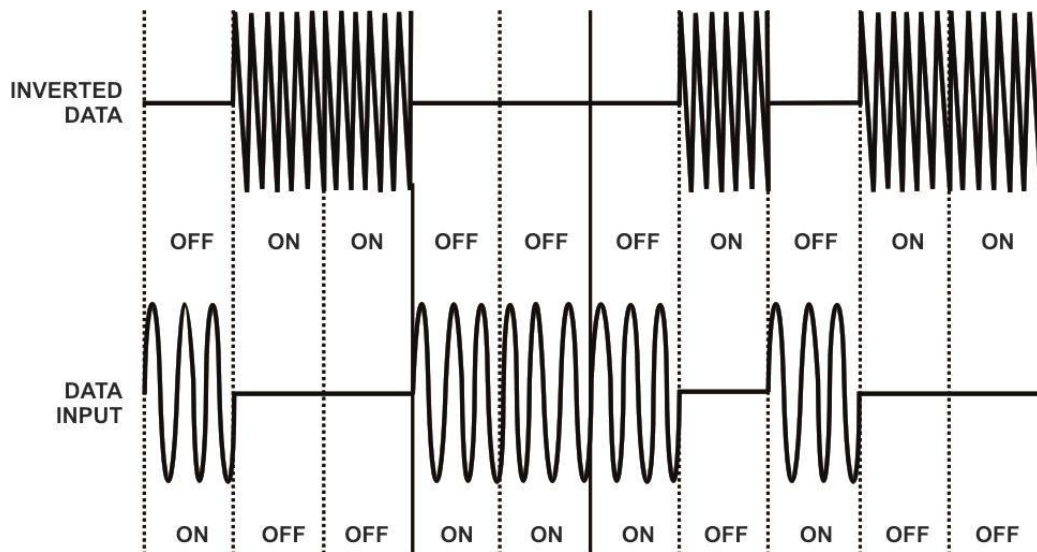


Fig. 3: FSK Modulation using ASK Modulation and Inverted Data Stream

Lastly, we have to sum the two ASK waveforms, to get a FSK wave.

Functional blocks required to generate the FSK signal are shown in Fig. 4. The two carriers have different frequencies & the digital data is inverted in one case.

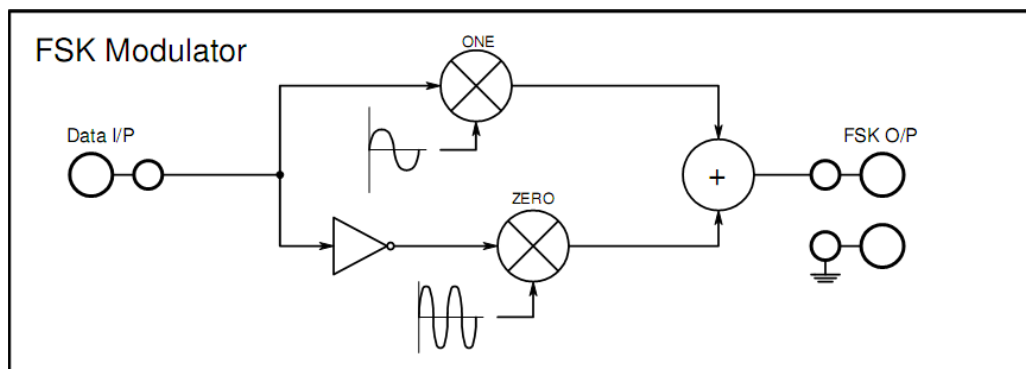


Fig. 4: FSK Modulation Functional Diagram

### Frequency Shift Keying De-Modulation

Method of FSK De-Modulation is also fairly simple. We need to design a tone decoder for one of the Carrier frequencies. For this Carrier frequency the tone decoder will output Logic '0' and for the other Carrier frequency it will output Logic '1'.

In practice this will be similar to ASK De-Modulation. We will have a VCO tuned to the Carrier frequency of Data '0', i.e. 300 KHz. For Data '1' the Carrier frequency will be around 165 KHz.

Now when VCO receives Carrier frequency for Data '0' it will output Logic '0' and when it will receive Carrier frequency for Data '1' it will output Logic '1'. Hence FSK Modulated Data Stream will be De-Modulated accordingly.

FSK De-Modulator Functional Blocks are shown in Fig. 5.

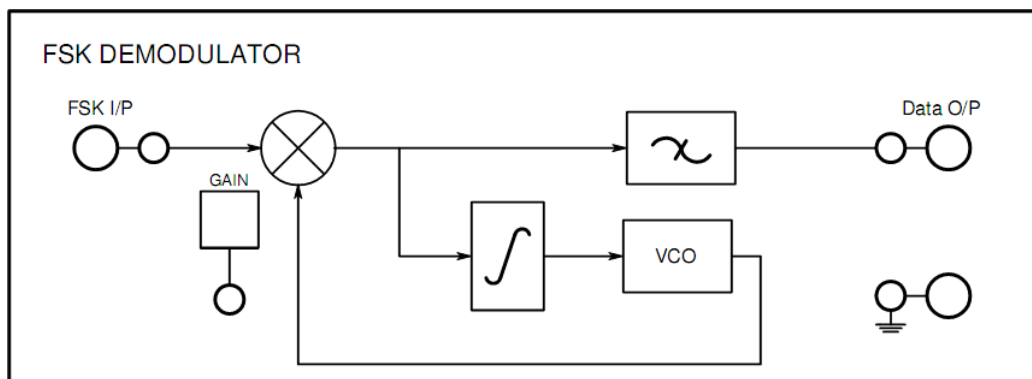


Fig. 5: FSK De-Modulator Functional Blocks

## Part A: FSK Modulation

### Instrument:

1. IT-4206 Trainer Board
2. 2mm Patch Cords
3. CRO

### Experimental Setup:

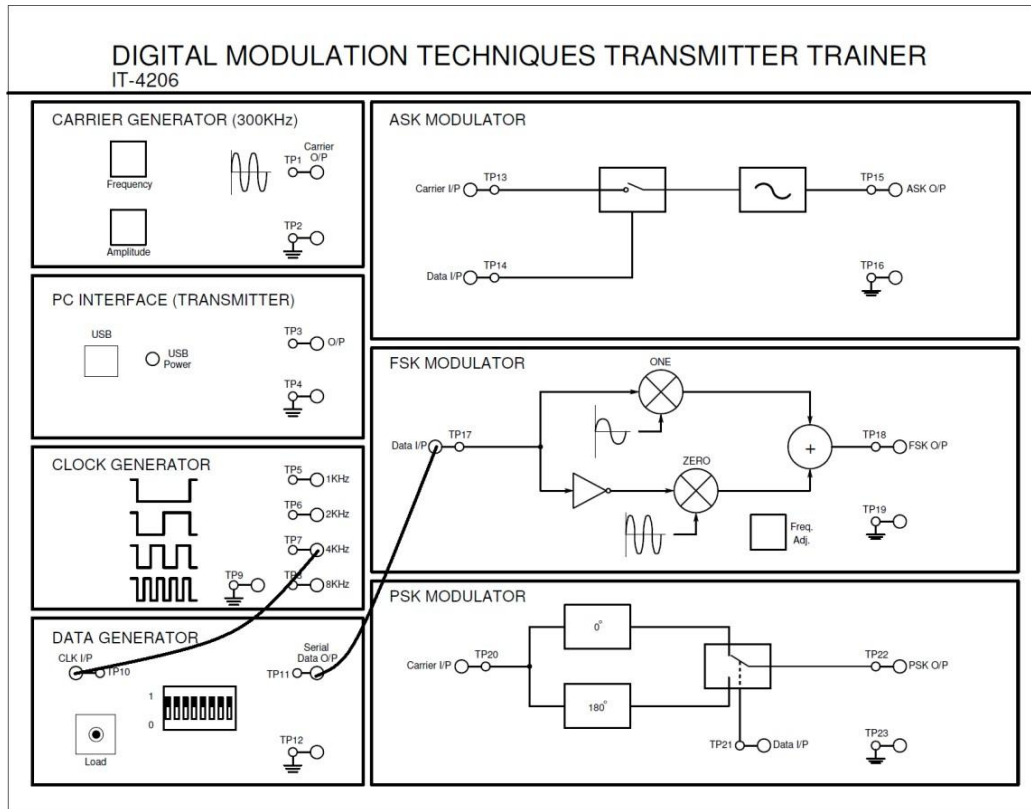


Fig. A.1

### Procedure:

1. Connect 4 KHz clock TP7 to the clock input of Data Generator. (Frequency of the clock can be changed according to the requirement)
2. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved.  
0 0 0 0 0 0 0
3. After setting the DIP switches press 'Load' switch.
4. Connect the Serial Data Output TP11 to the input of Data input of FSK Modulator TP17.
5. Carrier can be observed at FSK O/P test point TP18 using CH 1 of Oscilloscope
6. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved.  
1 1 1 1 1 1 1
7. After setting DIP switches press 'Load' switch.
8. Low frequency carrier can be observed at FSK O/P test point TP18 using CH1 of oscilloscope.

9. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved.  
1 0 0 1 1 1 0 1
10. After setting DIP switches press 'Load' switch
11. FSK Modulated waveform can be observed at FSK O/P test point TP18 using CH1 of oscilloscope.
12. Now Connect CH1 to Serial Data O/P test point TP11 and CH2 to FSK O/P Test point TP118 and verify the FSK Modulation waveform with respect to Data I/P.
13. Different Data Patterns can be obtained and experimented by changing DIP switches and pressing Load switch.

## Part B: FSK Demodulation

### Instrument:

1. IT-4206 Trainer Board
2. IT-4207 Trainer Board
3. 2mm Patch Cords
4. CRO

### Experimental Setup:

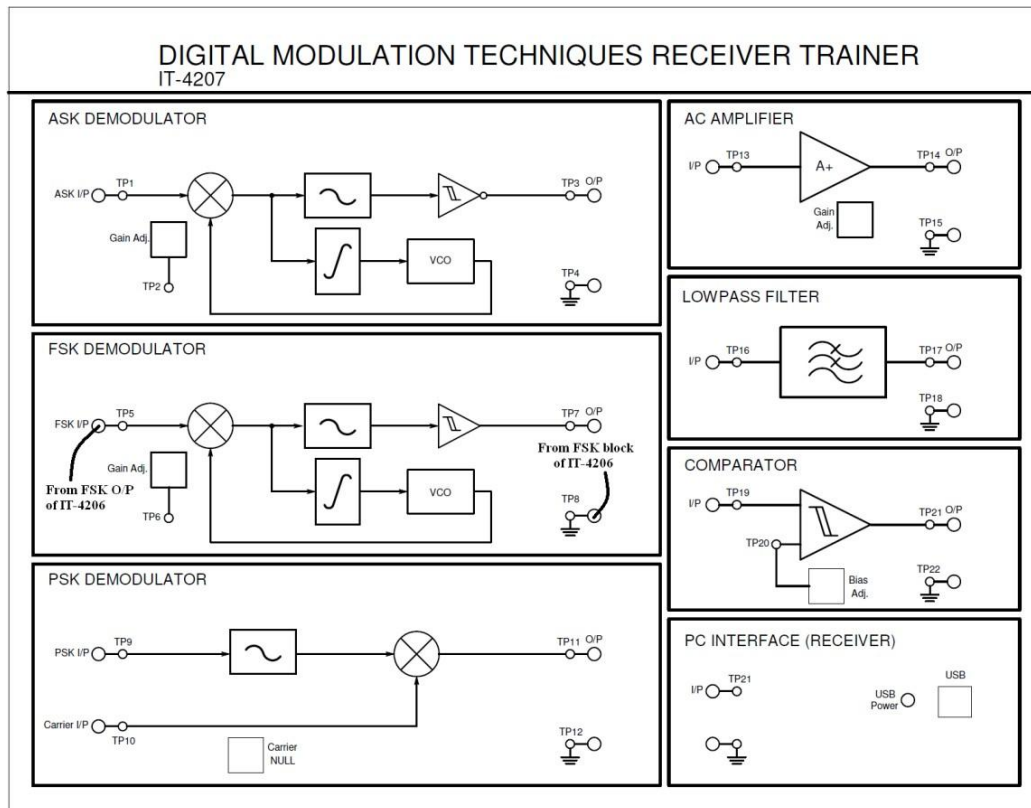


Fig B.1: Experimental Setup

### **Procedure**

1. Make connections as shown in Fig 4.1 using 2mm patch cords.
2. Now adjust the DIP Switches on IT-4206, starting from left, so that following data sequence is achieved. 1 0 0 1 1 1 0 1
3. Using IT-4206 Connect CH1 to Serial Data O/P test point TP11 and CH2 to FSK O/P Test point TP18 and verify the FSK Modulation waveform with respect to Data I/P.
4. Connect FSK O/P TP18 of IT-4206 to the input of FSK Demodulator TP5 of IT- 4207.
5. Connect the GND terminals of both modules using 2mm patch cord.
6. Now Connect CH1 to Serial Data O/P test point TP11 of IT-4206 and CH2 to Data O/P test point TP7 and verify that Transmitted Data is similar to that of Data received after FSK De-Modulation.
7. Different Data Patterns can be verified by changing DIP switches and pressing Load switch of IT-4206.



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Experiment No: 8 (Hardware)

Experiment Name: - Phase Shift-Keying

**Theoretical Background:**

Binary Phase Shift Keying (BPSK): BPSK (also sometimes called PRK, Phase Reversal Keying, or 2PSK) is the simplest form of phase shift keying (PSK). It uses two phases which are separated by  $180^\circ$  and so can also be termed 2-PSK.

It does not particularly matter exactly where the constellation points are positioned, as shown in Fig. 1, they are shown on the real axis, at  $0^\circ$  and  $180^\circ$ .

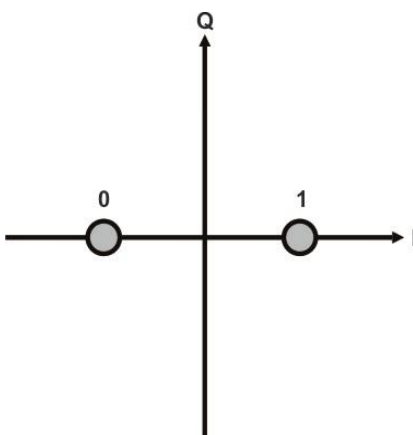


Fig. 1: BPSK Signal

This modulation is the most robust of all the PSKs since it takes the highest level of noise or distortion to make the demodulator reach an incorrect decision. It is, however, only able to modulate at 1 bit/symbol (as seen in the figure) and so is unsuitable for high data-rate applications when bandwidth is limited. Hence a BPSK Modulator will produce the waveform as shown in Fig. 2.

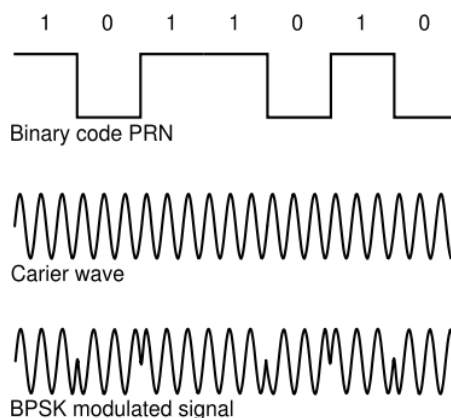


Fig. 2. PSK Modulated Waveform

PSK Modulator Functional Blocks are shown in Fig. 3.

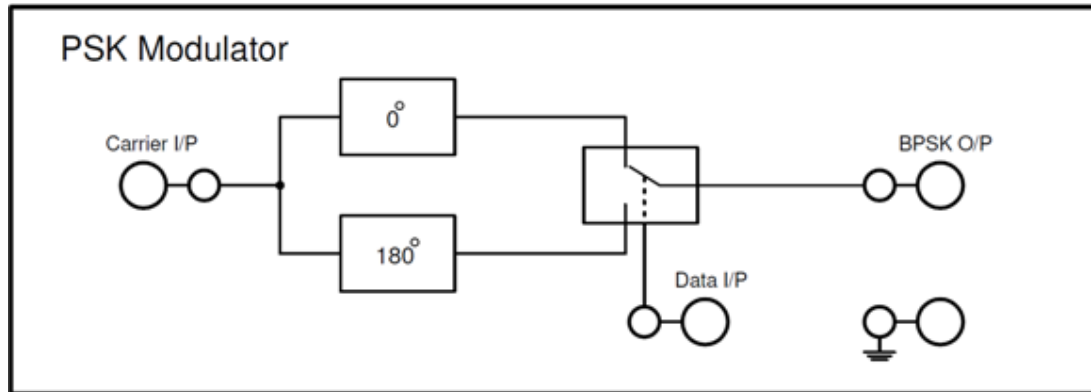


Fig. 3: PSK Functional Block

For implementation of PSK we will generate a Carrier and will also generate a synchronized inverted Carrier. Then these two phases of Carrier are fed to an Analog Switch. Data I/P will control the Analog Switch.

### Phase Shift Keying De-Modulation

The incoming PSK signal with  $0^\circ$  &  $180^\circ$  phase changes is first fed to the signal squarer, which multiplies the input signal by itself. The output of this block is a signal of twice the frequency with the frequency of the output doubled, the  $0^\circ$  &  $180^\circ$  phase changes are reflect as  $0^\circ$  &  $360^\circ$  phase changes. Since phase change of  $360^\circ$  is same as  $0^\circ$  phase change, it can be said that the signal squarer simply removes the phase transitions from the original PSK waveform.

The PLL block locks to the frequency of the signal square output & produces a clean square wave output of same frequency. To derive the square wave of same frequency as the incoming PSK signal, the PLL's output is divided by two in frequency domain is the divided by 2 circuit.

The following phase adjust circuit allows the phase of the digital signal to be adjusted with respect to the input PSK signal. Also its output controls the closing of an analog switch. When the output is high the switch closes & the original PSK signal is switched through the detector.

When the phases adjust block's output is low, the switch opens & the detector's output falls to 0 Volts. The demodulator output contains positive half cycles when the PSK input has one phase & only negative half cycles when the PSK input has another phase.

The phase adjust potentiometer is adjusted properly. The average level information of the demodulator output which contains the digital data information is extracted by the following low pass filter. The low pass filter output is too rounded to be used for digital processing. Therefore, it is 'Squared Up' by a voltage comparator.

PSK Demodulation Functional Blocks is shown in Fig. 4.

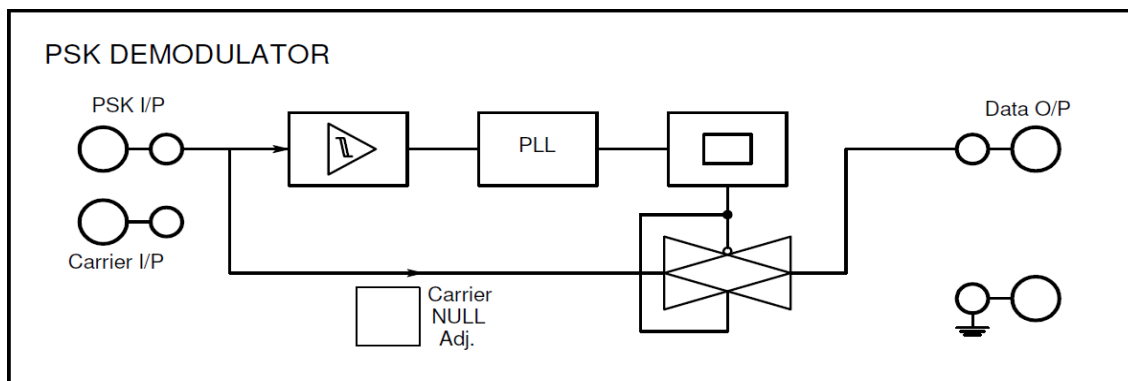


Fig. 4: PSK Functional Blocks

## Part A: PSK Modulation

### Instrument:

1. IT-4206 Trainer Board
2. 2mm Patch Cords
3. Digital Storage Oscilloscope

### Experimental Setup:

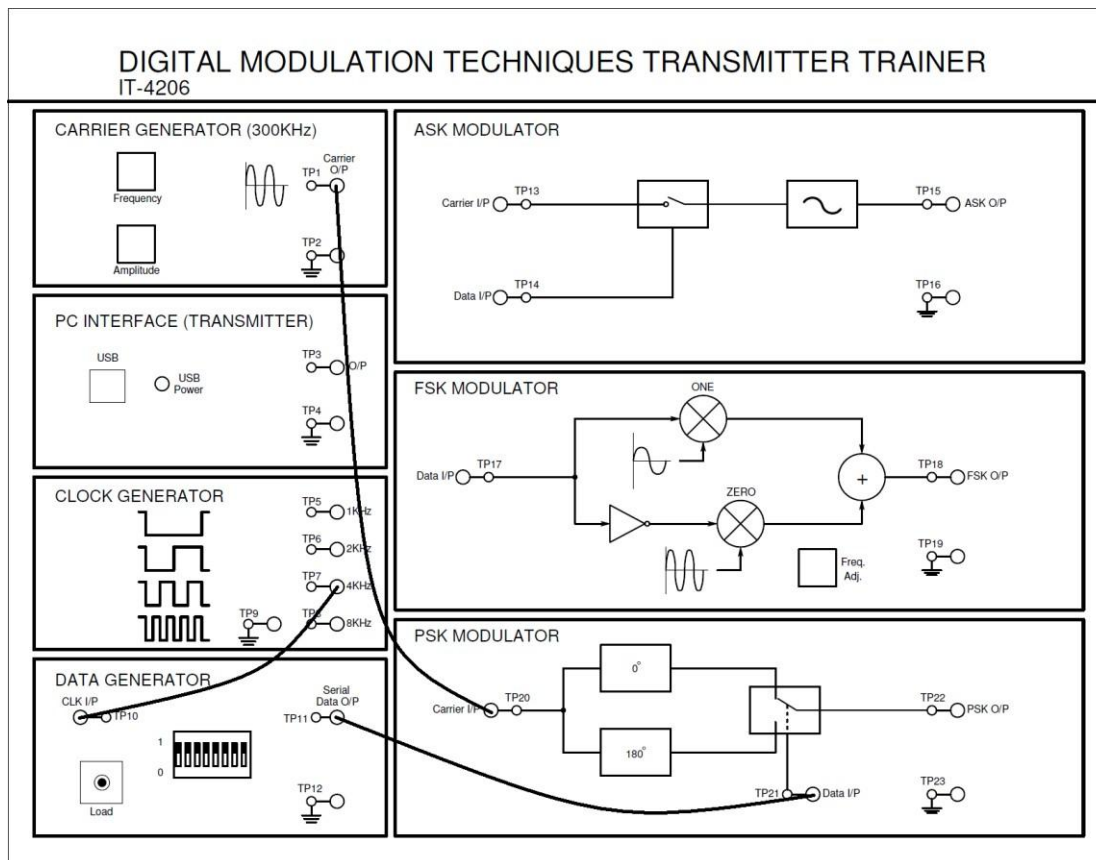


Fig 6.1: Experimental Setup

## Procedure

1. Connect Clock Generator output TP7 to the clock input of Data Generator TP11.
2. Now adjust the DIP Switches, starting from left, so that following data sequence is achieved. 1 0 0 1  
1 1 0 1
3. After setting DIP switches press 'Load' switch.
4. Connect Carrier O/P TP1 to Carrier I/P of PSK Modulator TP20.
5. Connect Serial Data output TP11 of Data Generator to the Data input of PSK Modulator TP21.
6. PSK Modulated waveform can be observed at PSK O/P test point TP22 using CH1 of DSO in Stop Mode.
7. Now Connect CH1 to Serial Data O/P test point TP11 and CH2 to PSK O/P Test point TP22 and verify the PSK Modulation waveform with respect to Data I/P.
8. Different Data Patterns can be obtained and experimented by changing DIP switches and pressing Load switch.

## Part B: PSK De-Modulation

### Instrument:

1. IT-4206 Trainer Board
2. IT-4207 Trainer Board
3. 2mm Patch Cords
4. Digital Storage Oscilloscope

### Experimental Setup:

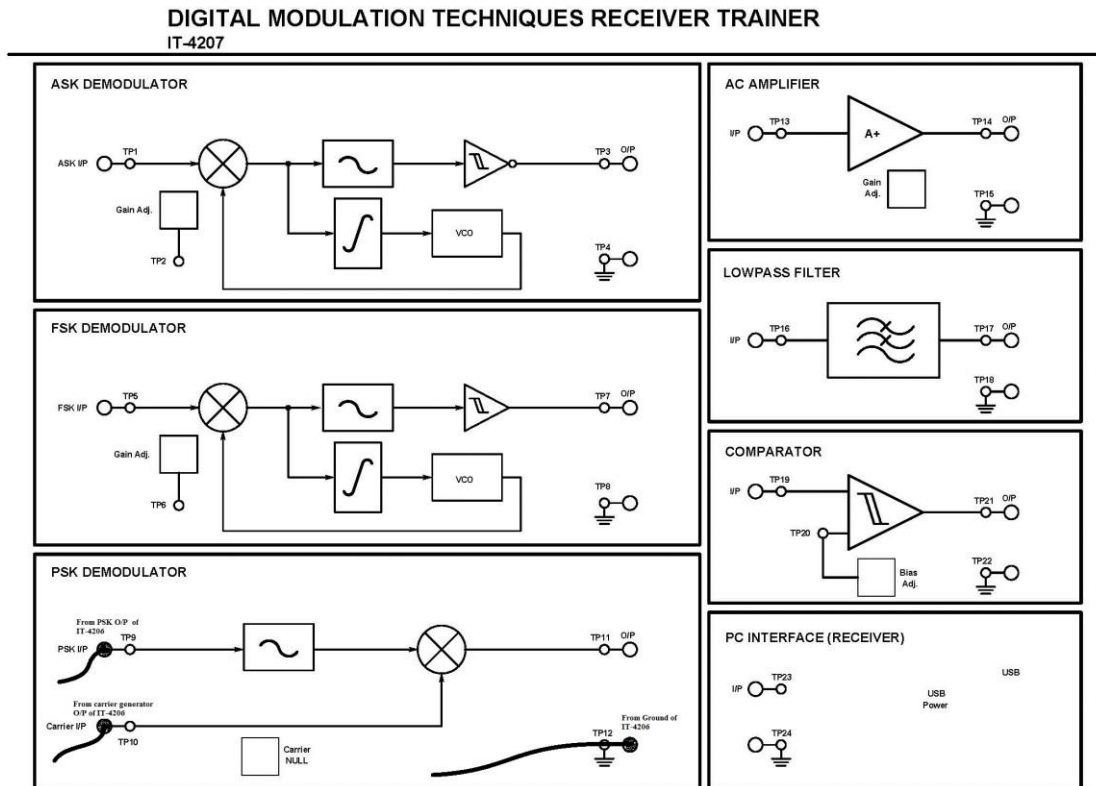


Fig. B.1: Experimental Setup

### Procedure

1. Make connections as shown in Fig. B.1 using 2mm patch cords.
2. Now adjust the DIP Switches on IT-4206, starting from left, so that following data sequence is achieved. 1 0 0 1 1 0 1.
3. Turn on the power of IT-4207 module.
4. Connect Carrier O/P TP1 of IT-4206 to Carrier I/P TP10 located in PSK Demodulator block of IT-4207.
5. Connect PSK O/P TP22 of IT-4206 to PSK I/P TP9 of IT-4207 and carrier at TP20 to the carrier input TP10 of PSK Demodulator.
6. Connect the GND terminals of both modules using 2mm patch cord.
7. Observe Signal at Data O/P test point TP11 of IT-4207 using CH2 of oscilloscope. Adjust the carrier null knob to get the data back.
8. Different Data Patterns can be verified by changing DIP switches and pressing Load switch of IT-4206.



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Experiment No: 6 (Software)

Experiment Name:- **Digital Pass band Modulation and Demodulation using MATLAB Simulink**

**Objective:**

The objective of this experiment is to study the digital modulation techniques such as OOK, ASK, FSK and PSK pass band modulation and demodulation using MATLAB Simulink. The transmitted binary sequence, the modulated and the demodulated signals will be observed in this experiment. An AWGN channel will be taken as the transmission medium.

**Theoretical Background:**

In a digital baseband system, the signals are transmitted directly over a pair of wires, coaxial cables, or optical fibers without any shift in frequencies of the signal. To remove the limitations of the antenna size required to transmit baseband digital signal, and for efficient use of the available frequency spectrum, passband transmission is suitable for transmission over radio links or satellites. Passband transmission is achieved by modulating a high frequency carrier signal by the baseband digital signal. In amplitude shift keying (ASK) and on-off keying (OOK), the amplitude of the carrier signal is varied with the amplitude of the digital signal. In frequency shift keying (FSK), the frequency and in phase shift keying (PSK), the phase of the carrier signal is varied with the amplitude of the digital baseband modulating signal respectively.

**Software and Device requirement:**

MATLAB (with Simulink)

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit); Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 Server (All Service Packs).

Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

**Problem Statement:**

(i) Draw all the models one-by-one in the MATLAB Simulink window.

(ii) Set the simulation time from the model window as below:

*Simulation Simulation parameters: Start time = 0.0 & Stop time = 10.0.*

(iii) Then simulate all the models and observe all the wave shapes at different points of the model.

**Procedure:**

- You can take the blocks from the Simulink Library Browser.
- The blocks that you will use in this experiment can be taken from the following mentioned library:

- (i) Bernoulli Binary Generator: Communication Blockset → Comm sources → Data → Sources → Bernoulli Binary Generator
- (ii) Sine Wave: Simulink → Sources → Sine wave
- (iii) Product: Simulink → Math operations → Product
- (iv) AWGN Channel: Communication Block Set → Channels → AWGN Channel
- (v) Digital Filter Design: Communication Blockset → Comm Filters → Filter Designs → Library Link → Digital Filter Design
- (vi) Spectrum Scope: Signal Processing Blockset → Signal Processing Sinks → Spectrum Scope
- (vii) Compare to Constant: Simulink → Logic & Bit Operations → Compare to Constant
- (viii) Scope: Simulink → Sinks → Scope
- (ix) Constant: Simulink → Sources → Constants
- (x) Abs: Simulink → Math operations → Abs
- (xi) Sum: Simulink → Math Operations → Sum

### Parameters to be used for different blocks-

- **Bernoulli Binary Generator:** *Probability of Zero = 0.5, Initial seed = 61, Sample time = 1 (for ASK & ON-OFF) and 1/10 (for FSK & PSK), Uncheck the 'Frame-based outputs' & 'Interpret Vector Parameters as one dimensional'.*
- **Spectrum Scope:**
  - ✓ Check 'Show Scope Properties' & 'Buffer input'. Set: *Buffer size: 128, Buffer overlap: 64 & Number of spectral averages: 2.*
  - ✓ Check 'Show axis properties'. Set: *Frequency unit: Hertz & Frequency range:  $[-F_s/2 \dots F_s/2]$ .*
  - ✓ After Simulation you will observe a window for each spectrum scope. To view the spectrum more clearly, set **Axes** → **Autoscale** from the spectrum window.
- **Sine wave:**
  - ✓ *For ON-OFF keying (Both modulator & demodulator):*  
*Sine type: Time based*  
*Amplitude: 1.0*  
*Frequency:  $2\pi \times 3$  rad/sec*  
*Sample time: 0.0001*
  - ✓ *For ASK:*  
*Sine type: Time based*  
*Amplitude: 5.0 (sine wave for '1'), 3.0 (sine wave for '0') & 1.0 (Sine wave in demodulator)*  
*Frequency:  $2\pi \times 3$  rad/sec*  
*Sample time: 0.0001*
  - ✓ *For FSK:*  
*Sine type: Time based*  
*Amplitude: 1.0*  
*Frequency:  $2\pi \times 20$  (carrier for '1' & sine wave 1) &  $2\pi \times 10$  (carrier for '0' & sine wave 0)*  
*Sample time: 0.0001*
  - ✓ *For PSK:*  
*Sine type: Time based*  
*Amplitude: 1.0*  
*Frequency:  $2\pi \times 10$  rad/sec*  
*Sample time: 1/5000*  
*Phase:  $\pi/2$  (carrier for '1' & sine wave 1) &  $-\pi/2$  (carrier for '0' & sine wave 0)*
- **Product:** *Number of inputs: 2 & Multiplication: Element wise*

- **AWGN Channel:** Initial seed: 67, Mode: Variance from mask & variance: 1.

Note: You can vary the noise power by varying variance

- **Digital Filter Design:**

Filter type: Lowpass

Design method: FIR → Equiripple

Filter order: Minimum order

Frequency Specifications: Units: Hz,  $F_s$ : 5000,  $F_{pass}=5$ ,  $F_{stop}=8$ .

Magnitude Specifications: Units: dB,  $A_{pass}=1$ ,  $A_{stop}=80$ .

Then click **Design Filter**. The magnitude response of the filter will be changed according to your specifications.

Note: You should choose your  $F_{pass}$  and  $F_{stop}$  frequencies depending on the frequency of your modulating signal.

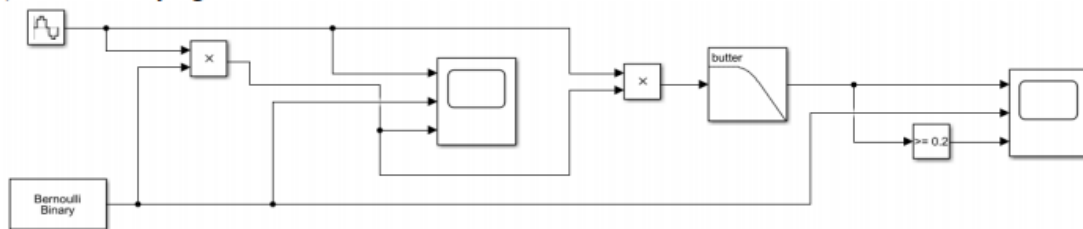
- **Compare to constant:** Operator:  $\geq$ . Constant: Depends on your system. This is the threshold value of the detector.
- **Scope:** Data History: Uncheck the 'Limit data points to last'.
- **Constant:** Constant value: 1
- **Abs:** Check the 'saturate on integer overflow' & 'Enable zero crossing detection'.
- **Sum:** List of signs:  $\pm$  (subtraction will be done).

Activate Wi  
Go to Settings

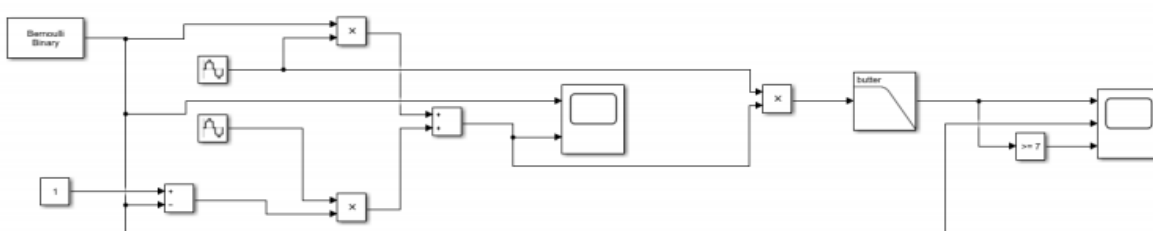
Note: To sum up two inputs, list of signs:  $++$

## Simulation Models-

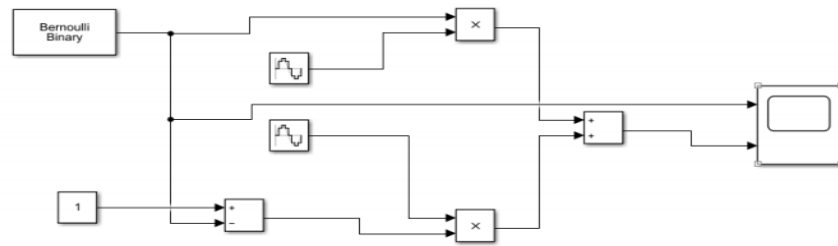
A) ON-OFF keying:



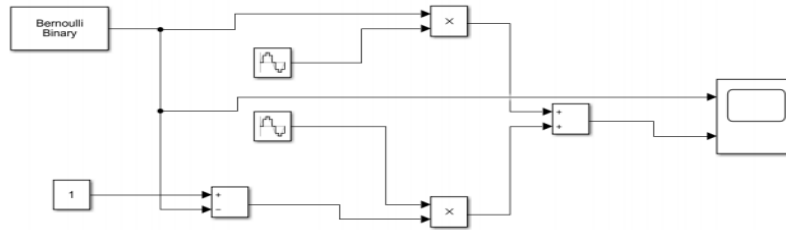
B) ASK:



C) FSK:



D) PSK:



### Lab Report:

1. Submit wave forms and frequency spectrums in printed form those you have observed in your lab.
2. Use an AWGN channel between the modulator and demodulator circuits for OOK and ASK. Describe the effects of noisy channel on Demodulation.
3. Construct demodulation circuits for FSK and PSK. Take help from theory lectures and internet.



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Experiment No: 7&8 (Software)

Experiment Name:- Digital Pass Band Modulation and Demodulation using MATLAB Simulink (FSK & PSK)

**Objective:**

The objective of this experiment is to study the digital modulation techniques such as FSK and PSK pass band modulation and demodulation.

**Problem Statement:**

At first, the binary sequence is transmitted, then the modulated and the demodulated signals will be observed in this experiment. An AWGN channel will be taken as the transmission medium.

**Software and Device requirement:**

MATLAB

Minimum PC specifications:

Windows/ Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit); Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 Server (All Service Packs).

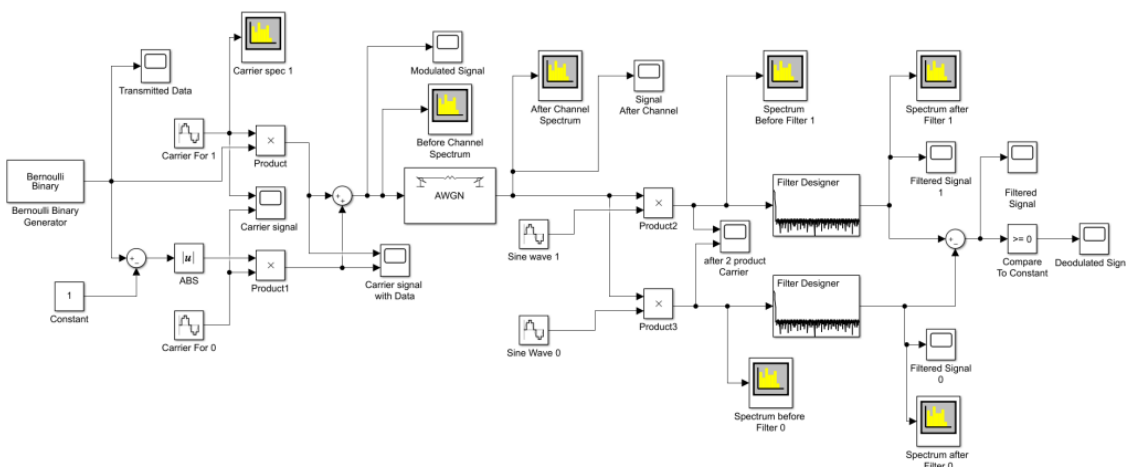
Ram: 2 GB

Processor: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU

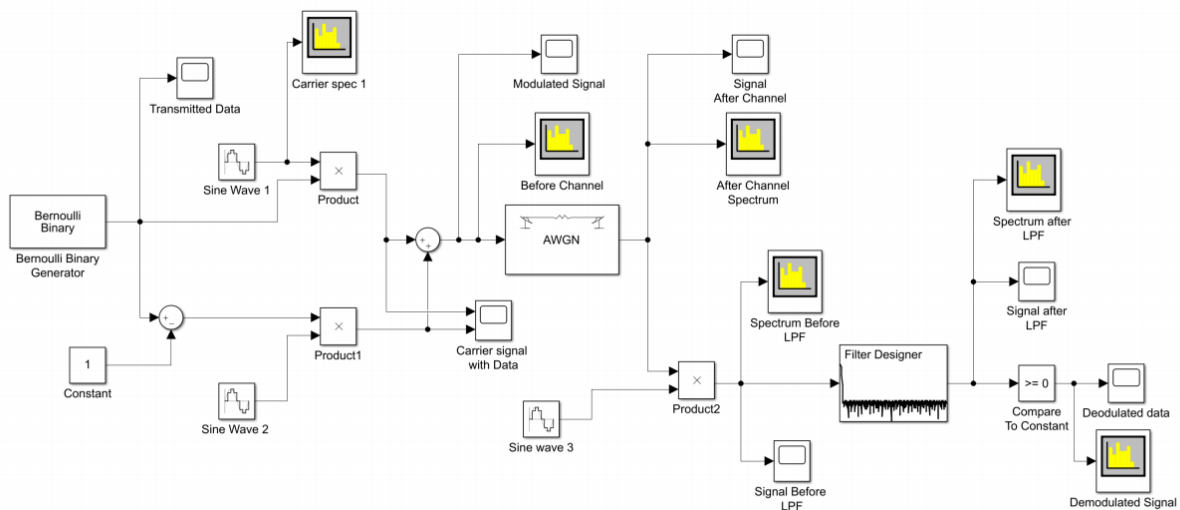
Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

**Circuit Diagram/ Experimental setup:**

**A) FSK:**



#### D) PSK:



#### Procedure:

You can take the blocks from the Simulink Library Browser.

The blocks that you will use in this experiment can be taken from the following mentioned library:

- Bernoulli Binary Generator: Communication Blockset→Comm sources→Data Sources→Bernoulli Binary Generator.
- Sine Wave: Simulink→Sources→Sine wave
- Product: Simulink→Math Operations→Product.
- AWGN Channel: Communication Block SetChannelsAWGN Channel.
- Digital Filter Design: Communication Blockset→Comm Filters→Filter Designs Library Link→Digital Filter Design.
- Spectrum Analyzer : DSP System ToolBox→Sinks→Spectrum Analyzer.

#### Parameters to be used for different blocks:

- Bernoulli Binary Generator: Probability of Zero = 0.5, Initial seed = 61, Sample time = 1(for ASK & ON-OFF) and 1/10 (for FSK & PSK), Uncheck the 'Frame-based outputs' & 'Interpret Vector Parameters as one dimensional'.
- Spectrum Analyzer : Spectrum Settings→Type : RMS , Trace Option →select "two-sided spectrum".
- Sine wave:

##### For FSK:

Sine type: Time based

Amplitude: 1.0

Frequency:  $2\pi \times 2000$  (carrier for '1' & sine wave 1) &  $2\pi \times 1000$  (carrier for '0' & sine wave 0).

Sample time: 1/5000

##### For PSK:

Sine type: Time based

Amplitude: 1.0

Frequency:  $2\pi \times 1000$  rad/sec

Sample time: 1/5000

- Product: Number of inputs: 2 & Multiplication: Element wise.
- AWGN Channel: Initial seed: 67, Mode: Variance from mask & variance: 1. (Note: You can vary the noise power by varying variance)
- Digital Filter Design:
  - Filter type: Low Pass.
  - Design method: FIR→Equiripple.
  - Filter order: Minimum order.
  - Frequency Specifications: Units: Hz, FS: 5000, FPass=15, FStop=55. (FSK)
  - Magnitude Specifications: Units: dB, Apass=1, Astop=80.
  - Then click Design Filter. The magnitude response of the filter will be changed according to your specifications.
  - Note: You should choose your Fpass and Fstop frequencies depending on the frequency of your modulating signal
- Compare to constant: Operator: >=. Constant: Depends on your system. This is the threshold value of the detector.
- Scope: Data History: Uncheck the 'Limit data points to last'.
- Constant: Constant value: 1
- Abs: Check the 'saturate on integer overflow' & 'Enable zero crossing detection'
- Sum: List of signs: |+- (subtraction will be done). (Note: To sum up two inputs, list of signs: |++)

#### **Lab report directions/ Questions/ Discussions/ Assignments:**

- A. Submit waveforms and frequency spectrums in printed form to those you have observed in your lab.
- B. Change FPass and FStop for all the filters to 100 Hz and 200 Hz respectively for both models. Then observe both the input and the output wave shapes and frequency spectrums for both models. Is there any discrepancy between the input and the output wave shapes? If so, explain why?
- C. Increase the variance in the AWGN channel block to 10 and 100. Is there any effect of this change on the output wave shapes and frequency spectrums? If any, what are the changes?
- D. "The received signal in analogue communication is distorted more by the channel noise than that of digital communication" – If you agree with this statement, then explain why?

## PROJECT

### Design and simulation of a communication link using AM

#### Expected Outcomes:

To have built communication links using existing AM modulation and demodulation blocks, constructed AM modulators using operational function blocks based on their mathematical expressions, and conducted simulations of the links and modulators, all in Simulink.

#### Detailed Requirements:

Use Simulink to design a communication link for AM audio broadcasting. The message signal is a mono audio signal although you may not be able to transmit the full audio frequency range that is normally required for high quality sound.

#### The specification for the link is as follows:

Required signal to noise ratio (SNR) at the demodulated audio output of the receiver: 40 dB for a 1 kHz message signal at 50% modulation ( $m = 0.5$ ).

- Carrier frequency: 1.35 MHz
- Maximum RF bandwidth available 9 kHz
- Channel noise power spectral density = -150dBm/Hz

Find out the following for Task-01:

1. What is the highest frequency of the message signal that can be transmitted without exceeding the specified RF bandwidth?
2. For this message frequency, save a time domain plot and a frequency domain plot showing the modulated RF output from the transmitter.
3. How much carrier power is required in order to achieve the required SNR? For this carrier power, how much power is there in each sideband for the  $m = 0.5$ ?
4. What is the SNR at the demodulated output if the frequency of the message signal is changed to the following frequencies?
  - 200 Hz
  - The highest frequency that can be transmitted without exceeding the specified RF bandwidth.
  - What is the SNR at the demodulated output if the modulation index  $m$  is increased to 1?
5. What happens if  $m > 1$ , if  $m = 1.5$ ? Compare the demodulated output from the receiver in the time domain and in the frequency domain for  $m = 1$  and  $m = 1.5$  and explain why a modulation index greater than 1 must be avoided in an AM link.

#### Prompts:

In order to complete the work required in the above, you will need to

- Generate baseband and carrier sinewave signals and AWGN noise
- Construct a channel model with constant loss and AWGN noise
- Construct an AM modulator with operational function blocks based on time-domain AM expression
- Construct a communications link using the built AM modulator, built channel model, and exiting AM demodulator block in Simulink.

## Design and simulation of communication links using PSK

### Expected Outcomes:

To have built communication links using existing PSK modulation and demodulation blocks, constructed PSK modulators using operational function blocks based on their mathematical expressions, and conducted simulations of both links and modulators, all in Simulink.

### Key Tasks:

- Generate baseband binary signals and carrier sinewave signals and AWGN noise
- Simulate and evaluate a communications link using BPSK with existing mod and de-mod blocks
- Simulate and evaluate a communications link using QPSK with existing mod and de-mod blocks
- Construct a BPSK modulator with operational function blocks based on the time-domain BPSK expression, and simulate and evaluate the BPSK modulator.

Find out the followings for Tasks 02 and 03:

1. You must measure BER against SNR or  $E_b/N_0$  and plot the performance curves according to the data obtained.
2. For the same noise level, in order to achieve a BER of  $10^{-5}$ , what is the signal power ratio of the BPSK and QPSK links?
3. Therefore, comment on BPSK and QPSK in terms of bandwidth efficiency and signal power required.
4. Show waveforms at different points of the link with different SNR (or  $E_b/N_0$ )
5. Show the constellations of the modulators

### Report:

#### For task-01 report must have the following:

Diagrams of communication links and modulators, simulated/calculated results and performances such as spectra (frequency domain), waveforms (time domain), analysis and discussions of results.

#### For task-02 and task-03 report must have the following:

Signals generated, link and modulator diagrams, simulation results including waveforms, evaluation of results, contrasting between BPSK and QPSK.

Lastly, include the references at the end to show the sources used to gather relevant information on the topic.